

CAPSTONE PROJECT 2

Final Report

Project Title

**Laundrify: Design and Development of an Intelligent Web-Based Laundry
Management Platform**

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Abstract

In today's world, digital transformation is being increasingly recognised as a vital component for the growth and survival of Small and Medium Enterprises, also known as SMEs in Malaysia (Rupeika-Apoga & Wendt, 2022). However, the service sector, specifically in the laundry businesses, often lags behind in adopting current world technologies. Many businesses and operators remain dependent on the usual manual workflows, such as handwritten logbooks and face-to-face communication which results in data redundancy, billing conflicts and lack of operational transparency (Rastogi & Tyagi, 2023). To address this gap, this project proposes a design and development of "Laundrify", which is an intelligent web-based online laundry management system. The main method in achieving this is through a quantitative requirement gathering process via a survey to study and validate the pain points of businesses and customers, followed by a structured system development lifecycle, to be implemented using Agile methods in Capstone 2. This proposed solution, Laundrify, integrates real-time order tracking, automatic invoices and a centralized dashboard for admins. The system was evaluated through a Technology Acceptance Model (TAM) survey (N=20) and structured Black Box Testing, which found strong perceived usefulness (4.76/5) and ease of use (4.9/5) among users, with all 11 test cases passing verification, confirming strong functional reliability across every tested module. The importance and significance of this project lie in its potential to be able to bridge the digital divide for all micro-SMEs, offering a scalable, user-friendly framework that enhances operational efficiency and user engagement in line with the current modern smart city initiatives. (Nazir & Khan, 2024)

Keywords: Online Laundry Management System, Automation, Digital Transformation, Service Design, SMEs, Web Application

Chapter 1

Introduction

1.1 Overview

In the current era of the industry, digital transformation has altered how business services are delivered and managed worldwide. While aspects such as banking and retail have adapted to automation, personal services such as the laundry industry often remain behind and reliant on traditional and manual workflows. This act of reliance on non-digital methods often results in traditional errors such as operational bottlenecks, data redundancy and a less connected customer experience.

To address these disadvantages, this project proposes the introduction of Laundrify, and it's in-depth journey of its design and development. Unlike complex and complicated enterprise systems often used by large franchises, Laundrify is specifically designed for Small and Medium Enterprises (SMEs) in Malaysia. This serves as a web-based hub that automates the orders, billing and provides real-time status tracking for the customers. By integrating these functions in Laundrify's web application, the project aims to show how low-cost digital innovations can significantly improve the operational performance and service reliability for micro-SMEs (Low et al., 2022; Malaysia Digital Economy Corporation, 2024).

1.2 Background

To fully understand and appreciate the real necessity of this project, it is imperative to study the current operational world of the laundry industry. This aspect is characterized by high volume, low profit margins and reliance on low-skilled labor. (Department of Statistics Malaysia, 2024) Consequently, efficiency is requirement for survival, rather than a metric of success. The laundry industry in Malaysia currently, operates primarily between two distinct

models which are the “Self Service” model which uses coins and QR as a form of payment. The other model is the “Full-Service” model which covers washing, drying and folding model.

Table 1.1 Operational Comparison Between Self-Service and Full-Service Laundry Models

Feature	Self-Service (Coin Laundry)	Full Service (Drop-Off)
Customer Role	Active: Customer loads, waits, and transfers clothes.	Passive: Customer drops off clothes and leaves.
Payment Method	Cash Kiosks, QR Pay and Tokens	Cash (Over counter) and Manual Transfer
Wait Time	Customer waits on-site	Customer returns later (Next day or hours later).
Staff Involvement	Minimal (Maintenance only)	High: Staff washes, dry, folds, and packs.
Tracking Method	Visual (Customer watches machine).	Manual: Paper logbooks & tags
Primary Pain Point	Machine availability (Occupied machines).	Transparency: <i>"Is my laundry ready yet?"</i>

1.2.1 Divergence of Modernization

In recent years, the Self-Service model of laundry has seen rapid modernization, in which it utilizes IoT-enabled washing and drying machines as well as cashless payment kiosks. However, the Full-Service sector, which caters to busy individuals that require washing, drying, folding, dry cleaning and ironing services remains operationally stagnant.

While Self-Service laundry centers offer speed, they still require the customer to remain onsite or return to the store once the cycle is completed. In contrast, Full-Service laundries offer the convenience of dropping the laundry at the store, following a “Drop and Go” concept.

However, this convenience is still being impacted by the manual processing of orders. (Rastogi & Tyagi, 2023).

1.2.2 Manual Workflow Analysis

To study and identify the specific points of conflict and failure, it is very necessary to deconstruct the operational cycle of the standard “Drop-Of” laundry services. Unlike the usual self-service model where customers can retain custody of their garments, the full-service model usually relies on a chain of custody that are maintained through physical artifacts.

- i. Intake phase (High Friction): The service starts off when a customer hands the garments over and the staff manually weighs them and transcribe details which involves the customer’s name, phone number, service type and special requests into a physical paper. A carbon-copy receipt is produced for the customer as well as the shop.

Operational weakness: The manual transcription is a source of data degradation in which, the handwriting illegibility or errors tend to lead to “broken communication loops”, which disables to store to contact customers (Suseno, 2024)

- ii. Processing Phase: The garments are tagged paper tickets to help staff members identify the ownership during the cleaning stages.

Operational weakness: Physical tags tend to detach or degrade in environments such as laundry services. If the tag goes missing, the garments become an “Orphan inventory” which leads to significant time spent by the staff to visually identify the garments to the description on the logbooks. (Parasuraman et al., 1988).

- iii. Estimation Phase When a customer wonders on when the garments will be ready for collection, the staff member must make a guess based on intuition.

Operational weakness: The human estimation is poor, and staff often fail to account for machine availability, existing backlogs, or even system malfunction. This leads to two separate negative outcomes which are over promising, or under promising. (Maister, 1984)

- iv. Retrieval Phase: Upon a customer's return to the store, the staff is forced to search through racks of completed garments to locate the customer's bundle.

Operational Weakness: This “search time” refers to a significant operational bottleneck. During peak periods, the time taken to physically locate the specific bundles reduces the store's efficiency and increases the queue times for customers, which impacts customers satisfaction. (Buell & Norton, 2011)



Figure 1.1 Breakdown of Manual Workflow Inefficiencies and Critical Failure Points.

1.2.3 Evolution of Consumer Psychology

The need for this project is further motivated by the shift in consumer expectations. The widespread adoption of the culture “On-Demand Economy” platforms such as Grab and Shopee has conditioned the way Malaysian consumers set their expectations. These expectations involve three key elements: Transparency, Immediacy and Auditability. In the current world, modern consumers often view “Operational Opacity” without knowing the status of their service which serves as a main source of anxiety (Buell & Norton, 2011). The current traditional and manual model forces each customer to bear the burden of tracking in

which they must initiate phone calls or visits to inquire on their status. This communication model is inefficient for the customers, especially the anxious individuals. (Smith, 2008)

1.3 Problem Statement

This project is motivated by the following problem statement:

Many Small and Medium Enterprises (SMEs) in the service sector often struggle to integrate digital automation into their internal practices, especially from the operational perspective. This often results in inefficient workflows, weak data integrity and mediocre suboptimal customer experience. Internal ecosystems in conventional laundry shops are still underutilized despite the potential of digital transformation, since there aren't enough easily available tools for non-technical business managers. (Kalo & Koppel, 2023)

The issue in Systemic Data Redundancy and Human Error, where data integrity is compromised by redundancy in a traditional manual environment. A customer's name and details of order are manually transcribed multiple times on different media such as the logbooks, receipts and tags. This act of repetition goes against the principle of "Single Source of Truth". These manual transcription points introduce non-zero probability of errors. If said, a staff member produces a receipt for 5kg but accidentally writes 4kg in the logbook, or miswrites a delivery date, the result will cost the business money, time and reputation. Manual transcription of service orders introduces a non-trivial probability of human error, particularly under high workload conditions, leading to inaccuracies, delayed processing, and reduced service reliability (Rupeika-Apoga & Wendt, 2022).

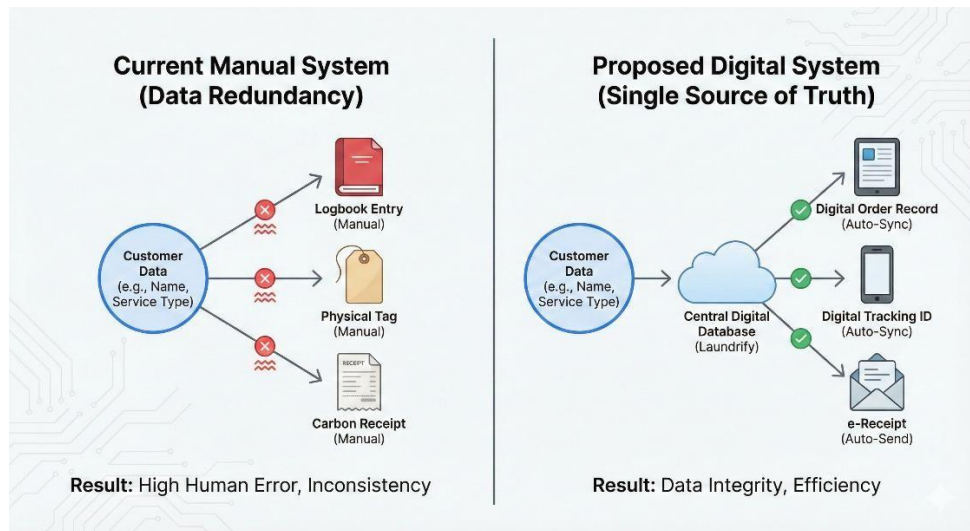


Figure 1.2 Comparison of Data Integrity: Manual Redundancy vs. Proposed Digital Single Source of Truth.

1.4 Objective

The main goal for this project is to design and develop a user-centric Laundry Management System, with insights into operational efficiency as a tool to mitigate the challenges faced by various business owners. Rather than focusing on financial aspects alone, a holistic service lifecycle approach will be integrated (Vargo & Lusch, 2008). The focus will be on improving internal data flow, status transparency and operational efficiency and consistency, with the business owner as well as the customer as the focal points of the system design.

To achieve this aim, three objectives must be acknowledged:

- i. Identifying the specific challenges, workflow bottlenecks, and digital obstacles faced by traditional laundry organizations through a quantitative requirement gathering process. (Wahid et al., 2021)
- ii. Designing and development of a fully functional web-based prototype, Laundrify, that integrates end-to-end order tracking, automated billing system, and real time status tracking based on identified challenges (Lizal & Dini, 2025)

iii. Evaluate effectiveness and the usability of the system using the Technology Acceptance Model (TAM) to ensure that the solution achieves a positive score in “Perceived Ease of use” for staff that are not tech-savvy (Davis, 1989).

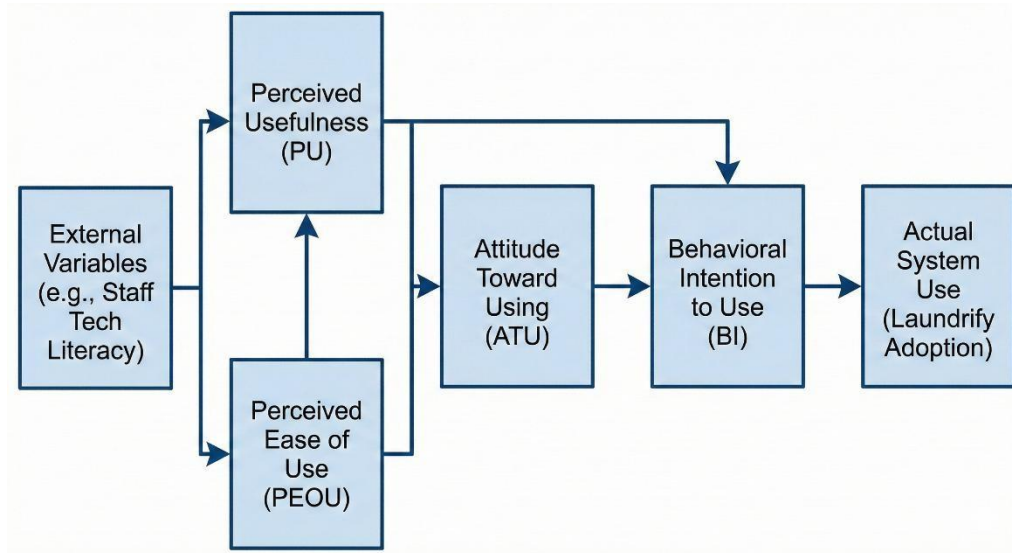


Figure 1.3 The Technology Acceptance Model (TAM) Framework (Adapted from Davis, 1989)

1.5 Scope

The scope of this project defines the boundaries, deliverables and limitations of the work to be done. This project focuses strictly on the development of a web-based Laundry Management System tailored to micro-SMEs, aiming to come up with an appropriate solution to resolve the challenges faced by business owners regarding manual and traditional ways of laundry management.

1.6 Structure of the Report

This proposal report is sectioned into four chapters, which details the complete lifecycle of the “Laundrify” project from problem analysis to the final evaluation. The structure of the report is outlined as follows:

- i. **Chapter 1:** Introduction is the chapter in which the foundation of the entire project is established. It represents the background of the chosen topic, defines problem statement regarding the current manual issues of the laundry industry and operations, and outlines the project's objectives, scope and overall significance. This first chapter serves as the roadmap for the development of "Laundrify"
- ii. **Chapter 2:** Literature review serves as the chapter that provides a theoretical framework for this project, by critically analyzing and studying the existing laundry management systems and gain feedback from current solutions and technologies implemented, including the MERN stack and automation algorithm systems. This review identifies the gaps in the current manual workflows, justifying the need for "Laundrify"
- iii. **Chapter 3:** Methodology is the chapter in which the research design and development of "Laundrify" is structured into three phases. Where, phase 1 details the requirement of gathering information via a survey, phase 2, outlining the design and development of the system with Agile methodology and finally phase 3, where the evaluation plan to validate the effectiveness of "Laundrify"
- iv. **Chapter 4:** Preliminary Results, Discussion and Conclusion, is where the outcome of this research is presented. It will detail the results of the requirement analysis, discuss the results of user evaluation while provide a proper conclusion of the significance of this proposal. It reiterates how the proposed solution, "Laundrify" aligns with the problems identified and provides a concluding overview of the expected project contribution in the laundry industry.

1.7 Summary

In Chapter 1, the birth of "Laundrify" has been proposed as a solution to operational inefficiencies and gaps in the current manual and traditional ways of the laundry service industry. The current reliance on manual logbooks, lack of real-time transparency and operational bottlenecks identify as critical issues affecting both service providers and service consumers. (Rastogi & Tyagi, 2023)

To address and overcome these issues, this project proposes the development of “Laundrify”, an Intelligent Web-Based Laundry Management System. This project aims to automate the workflows of laundry through a timer-based status system, which ensures accurate tracking of the laundry processes without the constant need for human intervention. By defining the project’s scope, objectives, and significance, this chapter has set a good starting stage for the subsequent phases of research. In chapter 2, Literature Review, we will further explore the current solutions available in the laundry industry and technical concepts to ground “Laundrify” in established research.

Chapter 2

Literature Review

2.1 Overview

The purpose of Chapter 2 is to be able to provide a comprehensive and critical analysis of the existing knowledge surrounding the digitalization inside the personal service industry, with a specific focus on the micro-SME laundry sector currently in Malaysia. While the first part of this proposal focused on the existence of operational inefficiencies through preliminary observations, this chapter aims to investigate the root causes of those errors and weaknesses, validate them against academic theory, and study on why previous technological interventions had failed tremendously to achieve a widespread adoption in the laundry industry. This section is not just a summary of existing tools and services, but it is a study and investigation into the intersection of Operational Management, Consumer Psychology and Information Systems. To provide a structured framework for this analysis, this chapter will be equally divided into five thematic sections.

- i. Landscape of Small and Medium Enterprises (SMEs) Digitalization: A study and analysis of the Malaysian's MyDigital framework, economic contribution of the services and the structural obstacles that prevent these SMEs from adopting the current technologies (Industry 4.0 standards)
- ii. Operational Dynamics of the Laundry World: A deep study of the "Service Lifecycle". This part contrasts Product-Dominant Logic with Service-Dominant Logic, to explain why generic software often fails. It also studies and analyses the risks of "Chain of Custody" failures and "Orphan Inventory"
- iii. Theoretical Framework: This section will be a justification for the usage of Technology Acceptance Model. In this section, we will explore the psychological aspects of Perceived Usefulness and Perceived Ease of Use, especially in the context of "Digital Immigrants"

- iv. Consumer Psychology in the On-Demand World: This section will focus on the use of Laundrify to explore the challenges faced by users when it comes to operational opacity, which results in heightened anxiety.
- v. Technological Architecture: This section would be a technical conversation which compares Native Mobile Applications vs Progressive Web Applications (PWA). This argues for the advantages of Cloud Saas (Software as a Service) framework for reducing the Total Cost of Ownership for SMEs.
- vi. Comparative Analysis of Existing Solutions: A systematic analysis of the current market solutions, from manual logbooks to enterprise ERPs, resulting in a “Research Gap” that will justify the development of Laundrify.

2.2 Landscape of SME Digitalization in Malaysia

Small and Medium Enterprises (SMEs) are often referred to as the backbones of the Malaysian economy in today’s world. According to the study at Department of Statistics Malaysia (2024), it is stated clearly that MSMEs contribute approximately 38.4% to the official national Gross Domestic Product (GDP) and employs close to 48% of the labor force. However, despite this great economic significance in the market, the individual productivity of these SMEs often lags global benchmarks due to the lack of technological adoption and integration. (Low et al., 2022)

2.2.1 “Mydigital” Initiatives and Industry 4.0

In recognizing this productivity gap, the Malaysian Government initiative and launched the MyDigital blueprint and the Malaysia Digital Economy Blueprint in hopes to transform Malaysia into a proper digitally-drive, high-income country by the year 2030. A key suite of this strategy is the digitalization of up to 875,000 MSEM countrywide.

However, academic sources suggest that the policy intent hasn’t fully translated into the reality for micro-enterprises. A phenomenon described as “Digital Dualism” refers to the contrast

between “High-Tech” manufacturing and “Low-Tech” service sector which means the contrast between robots and human manual labor (Malaysia Digital Economy Corporation, 2024). While incentives exist, they are often reserved to high-value industries, which leaves common services such as laundry under support. Thus, this project places itself within this gap, proposing an approach to digitalization that doesn’t rely on the government, but on accessible, low-cost technology. (Rupeika-Apoga & Wendt, 2022).

2.2.2 The Hybrid Disconnect in Service SMEs

The Hybrid Disconnect in Service SMEs A study of recent studies shows a paradox in SME behavior. It was observed that over 80% of the Malaysian service-based SMEs had integrated the digital platforms for advertisements. (Malaysia Digital Economy Corporation, 2024). These SMEs maintain active pages such as Facebook, Instagram and use WhatsApp Business as the line for marketing and are often listed on Google Maps. However, fewer than 1% integrate digital tools for internal operations such as inventory organization and financial auditing.

This creates a “Hybrid Disconnect” which means when the customer’s journey starts off digitally, by finding the “Nearest Laundry” on Google but is then transitioned into a traditional process upon entering the store, such as receiving a handwritten paper receipt. This action has severe implications for Data Integrity. Due to the operational data being recorded on a piece of paper, it cannot be in line with the digital marketing channels. Business owners are unable to send targeted messages to “Customers who haven’t visited in 30 days” because that data is trapped on a physical piece of paper in a logbook. It is argued that being able to bridge this gap is a very effective way to increase productivity growth in the SMEs sector (Malaysia Digital Economy Corporation, 2024).

2.2.3 Obstacles in Adopting” The “Digital Divide” in micro-SMEs

Despite the overwhelming benefits of digitalization, the adoption rate in Malaysian microSMEs remain extremely low. Malaysia Digital Economy Corporation (2024) describes this as the “Digital Dualism” in which the high-value manufacturing sectors are able to modernize rapidly while the service sector often lags behind. The literature identifies two main barriers in this aspect: Economic Constraints and Cognitive Load.

i. Economic Constraints: Total Cost of Ownership

For micro-enterprises, the decision to adopt these technological practices is primarily bound to financial aspects. Rastogi and Tyagi (2023) argue that the “Total Cost of Ownership” for existing laundry solutions are prohibitive. Platforms such as CleanCloud operate on a SaaS (Software as a Service) model which often charges \$100 USD per month. Malaysia Digital Economy Corporation (2024) also notes that for common Malaysian Laundry stores with an average cost of RM15, a monthly digitalization cost exceeding RM200 would be a significant cut from the net profit. Furthermore, Malaysia Digital Economy Corporation (2024) states that 6% of the micro-SMEs often lack the funds and money to adopt these digital tools and represents the primary reason for rejecting them, while preferring manual methods that have no capital expenditure (CAPEX).

ii. Cognitive Barriers: A Technophobia Effect

Beyond the worry of cost, the psychological barrier is equally significant. The demographic profile of many laundry owners in Malaysia in the current age often skew towards “Digital Immigrants” which are elder individuals who tend to be familiar with technology later in life (Venkatesh et al., 2003). Venkatesh et al. (2003) also states that in their Unified Theory of Acceptance, and Use of Technology, identify “Effort Expectancy” as a significant determinant of use. Current enterprise ERPs are most “feature-bloated” by offering many complicated and complex logistic modules, often

irrelevant to a single store. Venkatesh et al. (2003) found that if a specific digital tool requires a certain period of training, the abandonment rate for these SMEs spikes over 70%.

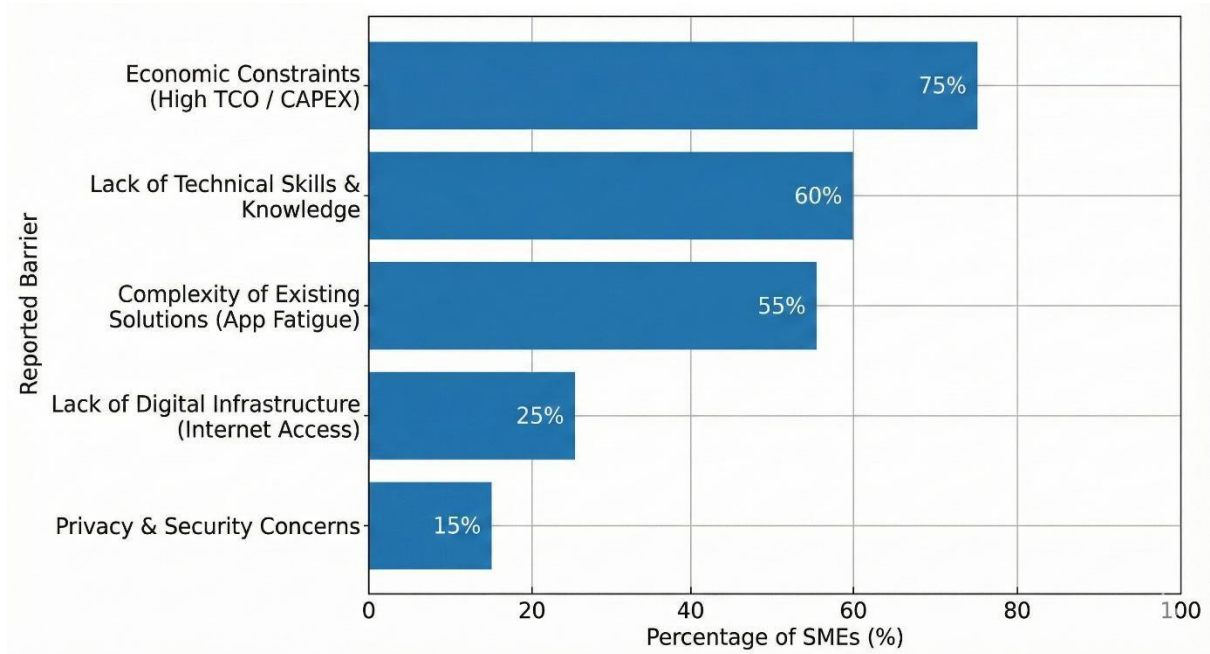


Figure 2.1: Reported Barriers to Digital Adoption Among Malaysian Micro-SMEs.

2.3 Operational Dynamics of the Laundry Service Industry

In efforts to design a viable software solution, one must first truly understand the niche operational constraints of the targeted industry, laundry. The laundry industry is often different from a standard retail, operating on logic that is often misunderstood by software developers.

2.3.1 Service-Dominant vs Product-Dominant Logic

In marketing theory, it distinguishes between two economic logics. Vargo and Lusch (2008) define Service-Dominant Logic (SDL) as when the value is co-created by a process over a span of time, while the Product-Dominant Logic (PDL) often focuses on the instantaneous exchange of products.

The most generic Point-of-Sale (POS) systems such as the StoreHub are built on ProductDominant-Logic, in which they can record a sale but are unable to track the process. Goduscheit and Faullant (2018) states that this mismatch is detrimental in the laundry industry. A POS system can record when a customer pays a fee of RM20 but is unable to answer one specific operational question: “*Is order #67 currently being washed in a washing machine or being dried in a drying machine?*”. This forces SMEs to use POS systems only for the final payment made by customers, while maintaining traditional methods of tracking the garments, such as manual logbooks. This extra effort is the main source of inefficiency.

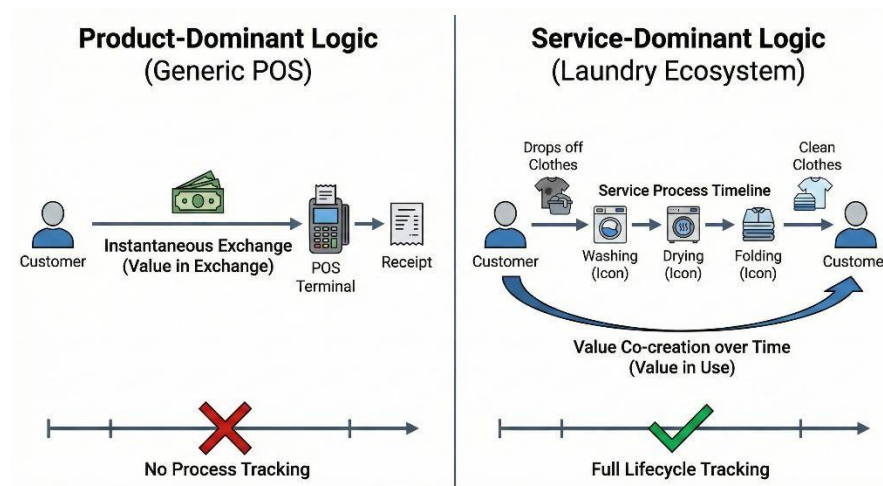


Figure 2.2 Contrast Between Product-Dominant Logic (Generic POS) and Service-Dominant Logic

2.3.2 The Chain of Custody and the “Orphan Inventory”

The fundamental “Drop-Off” model for laundry strictly relies on a rigorous Chain of Custody. Parasuraman et al (1988) states that “Reliability” holds the place of being the most important aspect of service quality. In the laundry industry, reliability is defined as an accurate return of a customer’s specific items.

However, in manual systems, custody is often maintained via Physical Artifacts, which consist of the manual logbook, a paper tag, and a carbon receipt. Unfortunately, physical artifacts are generally fragile as the paper tag often detaches or become illegible during the entire process and physical receipts frequently lose their small paper receipts.

When a customer's tag goes missing, the garments in the store now identify as "Orphan Inventory" in which items that have been processed but are unable to be linked back to a specific customer. Rastogi & Tyagi (2023) states that business that rely on these manual methods often lose up to 15-20% of their operation efficiency trying to overcome these "Orphan Inventory" issues by spending loads of time and cognitive efforts to visually inspect garments to match descriptions in logbooks. Digital systems are able to mitigate these risks by assigning unique Order IDs in database, that ensures the digital record remains as the "Single Source of Truth" (Buell & Norton, 2011).

2.4 Consumer Psychology in the current On-Demand World

The need for Laundrify is not only motivated by internal operation efficiency, but also the external consumer pressure. The rise of the platform economy in today's era has fundamentally altered consumer psychology in the service delivery aspect.

2.4.1 Operational Opacity and Consumer Anxiety

According to Buell and Norton (2011), they state that the term "Operational Opacity" is used to define a service process that are often hidden from the customers. They found out that opacity breeds the birth of anxiety and devaluations of the service in question. In a manual and traditional ways of the laundry industry, the process is considered a "Black-Box" in which the customer drops off their garments and have no visibility of the many processes until collection. (Maister, 1984)

Smith (2008) links this opacity directly to "Time Anxiety". Without any status updates to the customer, they tend to engage in "Monitoring Behavior", which generally means constant calls

to the shop to enquire if their garments are ready yet and questions regarding completion. This generates a cycle, in which a customer gets anxious which results in calls to the store. Through these actions, the staff get interrupted by the call, so they are bound to put their work on hold, which ends up in delayed work and decreased operational efficiency.

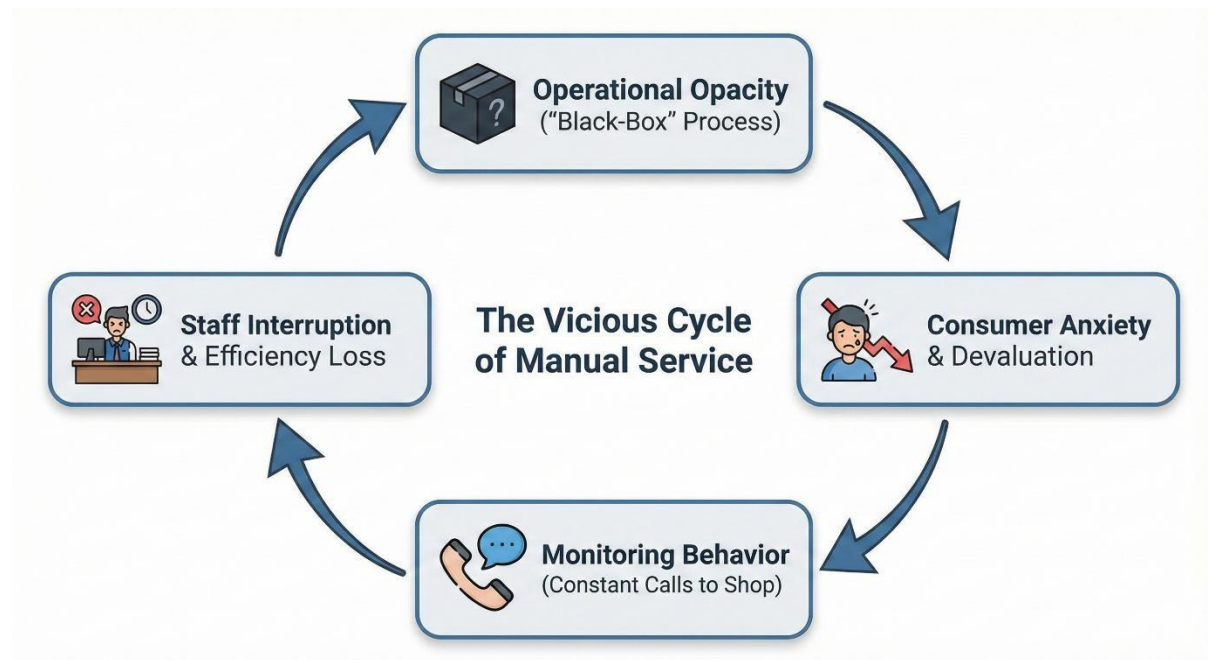


Figure 2.3: The "Vicious Cycle" of Operational Opacity and Consumer Anxiety in Manual Services

2.4.2 The Demand for Granular Transparency

Consumers conditioned by platforms like Grab now view Transparency as a proxy for Trust. Buell and Norton (2011) suggest that operational transparency increases perceived value. Applied to laundry, a Status Tracking Dashboard acts as psychological assurance, reducing anxiety and the need for constant calls.

2.5 Theoretical Framework: Technology Acceptance Model (TAM)

To ensure that Laundrify is not just technically sound, but also socially adopted by nontechnical-savvy staff, the design of the system is developed in line with the Technology Acceptance Model (TAM).

2.5.1 Overview of Technology Acceptance Model (TAM)

Proposed by Davis (1989), the Technology Acceptance Model (TAM) is known as one of the cited models for predicting user acceptance of information systems. While other models which are newer like UTAUT exist, TAM remains as the preferred framework when it comes to investigating simple applications in SME business context due to two aspects.

- i. Perceived Usefulness: “The degree to which a person believes that using a system would enhance his or her job performance” (Davis, 1989)
- ii. Perceived Ease of Use: “The degree to which a person believes that using a system would be free of effort” (Davis, 1989)

2.5.2 Applying Technology Acceptance Model (TAM) to mitigate “Digital Immigrants.”

In this study, the target users who are both the store owners and customers, are often “Digital Immigrants”, which are users who adopted technology in the later parts of their life (Venkatesh et al., 2003) In this demographic, the Perceived Ease of Use (PEOU) would be the dominant factor.

- i. Addressing the Perceived Usefulness: The system at hand must present an immediate value. For example, an “Automated Price Calculation” feature would directly address a pain point in mental math. If the user is able to see that “The system is able to do the math for me”, Perceived Usefulness increase tremendously.

ii. Addressing Perceived Ease of Use: If a user has to click ridiculous amounts of buttons to create an order, they will tend to revert back to manual ways of logbooks, which requires no clicks at all, as complexity often kills one's focus. Therefore, the UI design of Laundrify must be Minimalist and use Skeuomorphic design cues such as real-world recognized icons to reduce the cognitive load of the user. (Davis, 1989)

2.6 Technical Architecture: Web vs Native

In the development of a digital solution for SMEs in the current world, the choice of platform architecture is an extremely critical strategic decision, whether the choice be a Native Mobile Application, commonly on iOS or Android, or a Web-based Application (SaaS). A review of the technical portion strongly supports the usage of a Web Based approach for this project.

2.6.1 The Economics of Progressive Web Applications (PWAs)

Nagarajan and Rajan (2024) notes that the Progressive Web Applications (PWA) generally cost way lesser, approximately 30-70% lesser than to develop and maintain native applications. Native development requires two separate codebases which are Swift for iOS and Kotlin for Android. For the Laundrify project which aims for a low-cost product for SMEs, this dual development would not be efficient and would be costly. A web application utilizes one codebase, which consists of HTML, CSS and PHP, that is able to run on any device with a browser available, ensuring universal compatibility

2.6.2 The Issue of “App Fatigue”

Nikou and Economides (2020) highlights that the issue of “App Fatigue” is on the rise. Modern users these days are increasingly reluctant to download permanent applications for services that they require occasionally.

- i. From the Customer's Perspective: For users who visits a laundry shop perhaps once every two weeks, they would be unlikely to browse the Application store, wait for installation, and create an account just for a receipt. Not to mention that at times, certain applications would take up storage spaces in the device.
- ii. The Web's Advantage: A web application, however, allows for Zero-Friction Access, in which users simply scan QR codes, or at the click of a single link where the browser opens immediately without the use of installation and storage space. This "low barrier to entry would be crucial in maximizing the adoption of users.

2.7 Comparative Analysis of Existing Solutions

Finally, to identify the "Research Gap" in this study, a systematic comparative analysis of the current solutions in the market was conducted. The market is divided into three distinct tiers, which are Manual Methods, Generic POS and Enterprise ERPs.

2.7.1 Tier 1: The Manual System

The manual system holds its place as the default state for about 90% of SMEs.

- i. Mechanism: Staff writes the details of an order in a physical book, and handwrites a paper receipt for customers
- ii. Critical Weakness: This practice opens doors for lack of data redundancy, in which if the book was damaged, all records of customers information, debts and inventories are completely wiped out. (Suseno, 2024)

2.7.2 Tier 2: Generic Point-of-Sale

This method is highly used in tablet-based systems, specifically in retail and F&B sections.

- i. Mechanism: A tablet is presented as transaction-focused software, designed to complete purchases and sales
- ii. Critical Weakness: While this may be good practice for payments, it lacks the Lifecycle Logic, in which it is required for laundry. It records a “Sale” but is unable to record the transitions of that sale from “Dirty” to “Clean”. This would be a solution to financial issues, but would be inefficient for operational conflicts. (Rupeika-Apoga & Wendt, 2022)

2.7.3 Tier 3: Enterprise ERPs

The method is widely known for being sophisticated cloud platforms designed for larger franchises.

- i. Mechanism: Presents itself as a full-suite ecosystem, complete with transportation apps, customer apps and POS
- ii. Critical Weakness: This method is very costly as subscription fees are often priced at high amounts which are sustainable for large franchises, but not viable for smaller micro-SMEs. (Rastogi & Tyagi, 2023).

Table 2.1: Comparative Analysis of Existing Market Solutions vs. The Proposed "Laundrify" System.

Feature	Manual Logbook (Traditional)	Generic POS	Enterprise ERP	Laundrify (Proposed)
---------	------------------------------------	-------------	-------------------	-------------------------

Cost	Negligible	Moderate	Very High	Low (Done by web hosting)
Lifecycle Tracking	None	None	Advanced (Complicated)	Core Feature
Transparency	Low (Done through a phone call)	Low (Paper)	High	High (User Dashboard)
Ease of Use	High	Medium	Low (Complex systems)	High (User friendly and simple to use)

2.8 Summary

This section of the proposal has provided a critical, multi-dimensional analysis of the laundry management framework. The literature review has confirmed that the micro-SMEs sector is currently caught up in a “Digital Trap” in which they are stuck between inefficient manual logbook practices, and inaccessible costly enterprise software. The analysis of Consumer Psychology is proof of the urgent need for transparency for users to decrease any risks of anxiety and inconvenience. (Buell & Norton, 2011) The Technology Acceptance Model (TAM) provides the blueprint for the development of a system that goes in line with the limitations of “Digital Immigrants.” (Davis, 1989). Overall, the technical review of the laundry industry confirms that a Cloud-Based Web Application Architecture is an optimal solution in all aspects, by balancing the cost, accessibility and functionality. These studies and insights reports inform the methodology outlined below in Chapter 3, Methodology.

Chapter 3

Methodology

3.1 Overview

This chapter showcases how the three-phase methodology which was proposed in the Capstone 1 proposal was carried out during the Capstone 2 project. Phase 1, which was Challenge Identification was completed in the previous semester and its findings are carried forward and explain and summarized in section 4.2. This chapter focuses on how that information were translated into Phase 2, which is System Design and Development and Phase 3, Evaluation, which includes the points at which the actual implementation diverged from what the original proposal was, why and what factors and constraints shaped each decision.

3.2 Phase 1: Challenge Identification

Phase 1 was carried out as outlined in the Capstone 1 proposal. It used a mixed-methods approach to gather requirements, combining a comparative review of existing laundry management solutions (Chapter 2) with a quantitative Google Form survey involving 20 respondents. This survey was handed out to various students through WhatsApp primarily and Discord. Purposive sampling targeted hostel residents specifically because they are high-frequency consumers of laundry services and therefore the most reliable source of first-hand pain points.

The survey instrument, participant details, and overall procedure are fully documented in the Capstone 1 report, which is reproduced in Appendix A, and these were not changed for Capstone 2. The complete findings from this phase, including the Functional Requirement Specification (FR-01 to FR-04), are presented in Section 4.2 and serve as the empirical foundation for all design decisions discussed in this chapter.

3.3 Phase 2: System Design and Development

3.3.1 Development Methodology

As proposed beforehand, Laundrify was developed using the Agile Software Development Life Cycle, in which is structured as a lightweight Scrum of three sprints. This was chosen because an individual developer capstone project benefits from short, testable iterations rather than a single up-front design phase which is a decision validated in practice, since some defects were only discoverable once real navigation and testing could be exercised end-to-end and not from static code review alone, which is talked about in detail in Section 3.4.2

Table 3.1: Sprint Breakdown for the Laundrify development.

Sprint	Duration	Focus
Sprint 1	Week 1-2	Backend setup, schema, auth
Sprint 2	Week 3-4	Frontend, Dashboards
Sprint 3	Week 5-6	Integration, workflow engine, deployment
Phase 3a	Week 7-8	Black Box Testing, defect fixing
Phase 3b	Week 9-10	TAM Survey distribution and collection
Phase 3c	Week 11-12	Screenshot capturing, report writing

The MERN stack which consist of MongoDB, Express.js, React.js and Node.js was used as proposed in Capstone 1, on the justification that a single Javascript-based stack across client and server reduces context-switching overhead for an individual developer while remaining free-tier deployable, directly supporting the Total Cost of Ownership argument made in literature review.

3.3.2 System Architecture

Laundrify strictly follows a three-client-server architecture, in which a React single-page application communicates with an Express/node REST API which persists data to MongoDB Atlas and calls the Brevo API for transactional email. Authentication is stateless, which uses JWT bearer tokens attached to the Authorization header of every request, which reduces the chance of server-side session storage and keeps the backend scalable on Render's free tier.

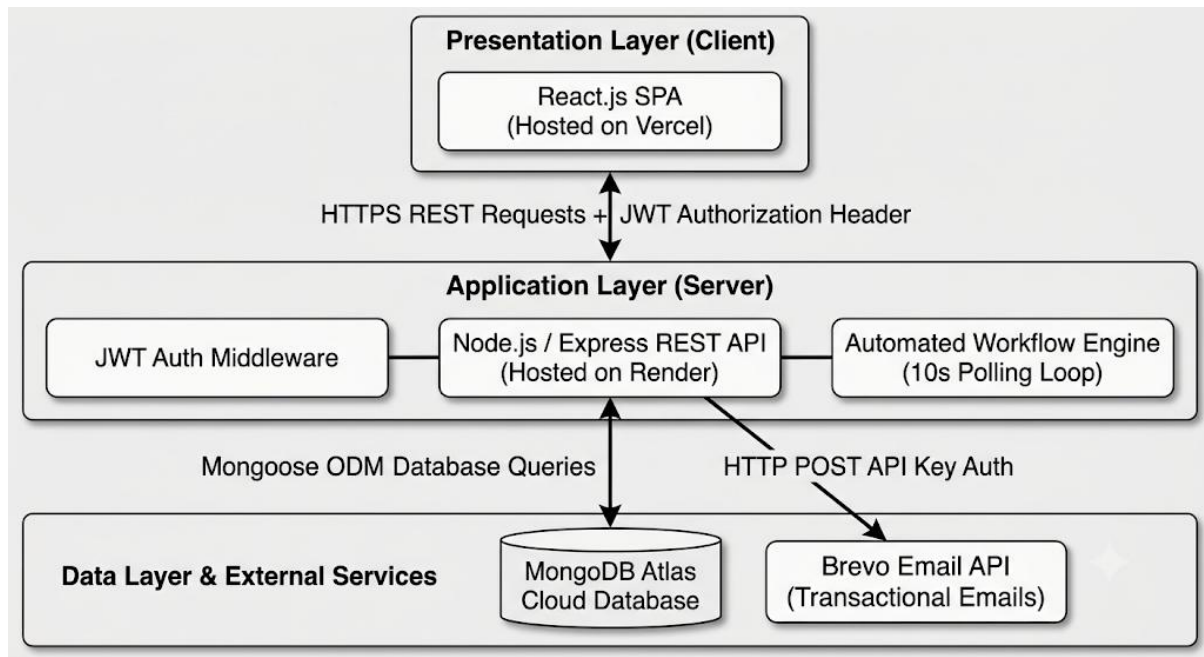


Figure 3.1: System Architecture Diagram.

3.3.3 Database Design

The database design used in Laundrify contains of four collections, with each mapping directly to a functional requirement which was identified in Phase 1

- Users: name, email address, bcrypt-hashed password. Password hashing addressed a security requirement in which credentials are never stored in plaintext at rest.
- Orders: customer details, service type, clothes type, weight, wash style, add-ons (such as drying, ironing and folding), price breakdown, status, payment status, feedback rating, and live timestamps. This implements FR-01, live status and FR-02, automated transitions

- Inventory: item name, quantity, unit, minimum stock threshold, category. This system supports low-stock alerts as well
- Staff: name, email address, phone number, shift and active status in which are all handled by error handling

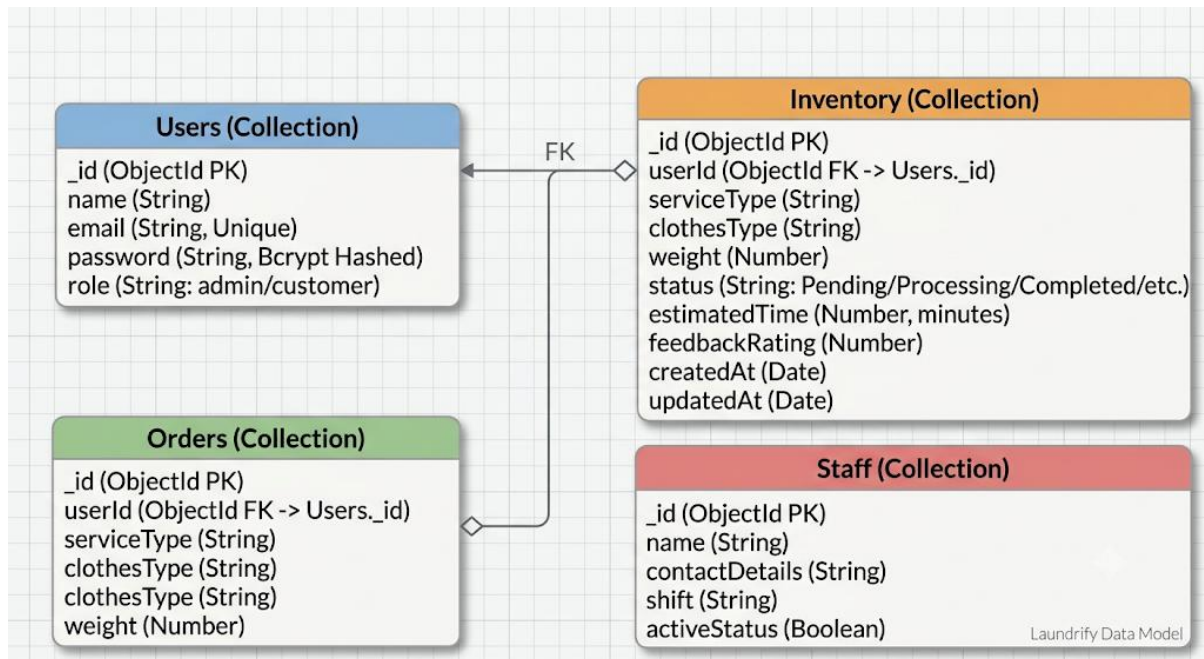


Figure 3.2: Database Design Structure.

3.3.4 Core Feature Implementation

Automated Workflow Engine (FR-02)

This proposal's deterministic timer login was implemented as a server-side interval that oversees all orders in the "Processing" status every 10 seconds. When the elapsed time since the last status update exceeds the order's specific estimated duration, the order automatically transitions from "Processing" to "Completed" and a completion email is triggered, which sends to the Customer's email address, which removes the need for staff to manually track individual loads and contact them. The 10-second timing interval was performance trade-off in which often times, customers see near-real-time updates but coarse enough not to place unnecessary real load on the database.

```

setInterval(async () => {
  try {
    const processingOrders = await Order.find({ status: 'Processing' });
    for (let order of processingOrders) {
      const processingTime = Date.now() - new Date(order.updatedAt).getTime();
      const estimatedMs = (order.estimatedTime || 30) * 60 * 1000;
      if (processingTime > estimatedMs) {
        order.status = 'Completed';
        order.completedAt = new Date();
        await order.save();
        console.log(`✅ Order #${order._id.toString().slice(-4)} Auto-Completed!`);
        await sendOrderCompleteEmail(order);
      }
    }
  } catch (error) {
    console.error('❌ Automation error:', error);
  }
}, 10000);

```

Figure 3.3: Code Snippet for the Automated Workflow Engine

Role-Based Access Control (FR-04)

RBAC is integrated at two layers rather than one, which acts as a direct security-by-design response to a defect found during testing. At the UI level, the navigation array rendered to each user is built conditionally on *user.role*, which means a customer account never receives a Settings link in the first place, which was removed entirely and not hidden with CSS. At the page-resolution layer, App.js's *renderPage()* switch statement simply has no case for admin-only pages under the customer role, meaning components such as StaffManagement, InventoryManagement, and Settings are never referenced on that code path and cannot be rendered regardless of the *currentPage* state. This is exclusion by construction rather than a check that runs after the fact and redirects. This second layer exists because testing showed that UI-only concealment (hiding the nav link) does not stop a customer from reaching an admin page by manipulating client-side state directly (Section 3.4.2), which is a reminder that presentation-layer hiding alone is a usability nicety, not a security boundary.


```

const renderPage = () => {
  if (!token || !user) {
    return isLogin ? (
      <Login onLogin={handleAuth} onSwitch={() => setIsLogin(false)} />
    ) : (
      <Register onRegister={handleAuth} onSwitch={() => setIsLogin(true)} />
    );
  }

  const commonProps = {
    user,
    onLogout: handleLogout,
    token,
    onNavigate: handleNavigate,
    currentPage,
    setUser,
    mobileMenuOpen,
    setMobileMenuOpen
  };

  if (user.role === 'admin') {
    switch(currentPage) {
      case 'dashboard': return <Dashboard {...commonProps} />;
      case 'inventory': return <InventoryManagement {...commonProps} />;
      case 'staff': return <StaffManagement {...commonProps} />;
      case 'history': return <OrderHistory {...commonProps} />;
      case 'settings': return <Settings {...commonProps} />;
      case 'profile': return <Profile {...commonProps} />;
      default: return <Dashboard {...commonProps} />;
    }
  }

  switch(currentPage) {
    case 'dashboard': return <CustomerDashboard {...commonProps} />;
    case 'history': return <OrderHistory {...commonProps} />;
    case 'profile': return <Profile {...commonProps} />;
    default: return <CustomerDashboard {...commonProps} />;
  }
};

```

Figure 3.4: Code Snippet for Role-Based Page Resolution (App.js)

Email Notification Service (FR-03)

In Capstone 1, the original proposal mentioned SMTP/Nodemailer as the primary notification sending mechanism. However, in practice, this had to be revised during deployment as Render's free tier blocks outbound SMTP connections, which means that the normal Gmail SMTP worked perfectly locally but failed in production. This is documented as a genuine, evidence-based deviation from the original proposal plan, rather than an oversight

Table 3.2: Email Provider Evaluation and Final Selection

Provider	Outcome	Reasoning
----------	---------	-----------

Gmail SMTP (Nodemailer)	Abandoned	Render's free tier blocks outbound SMTP connections
Resend	Abandoned	IPv6 connectivity issues within Render's infrastructure
Mailgun	Abandoned	Sandbox mode restricts sending to pre-authorised recipients, which is contradictory for Laundrify
Brevo	Adopted	API-based delivery, bypassing Render's restrictions, Free

```

Tabnine | Edit | Test | Explain | Document
async function sendOrderCompleteEmail(order) {
  try {
    const response = await fetch('https://api.brevo.com/v3/smtp/email', {
      method: 'POST',
      headers: {
        'api-key': process.env.BREVO_API_KEY,
        'Content-Type': 'application/json',
        'Accept': 'application/json',
      },
      body: JSON.stringify({
        sender: {
          name: process.env.FROM_NAME || 'Laundrify',
          email: process.env.FROM_EMAIL,
        },
        to: [{ email: order.customerEmail, name: order.customerName }],
        subject: `Order #${order._id.toString().slice(-4)} - Ready for Pickup`,
        htmlContent: `


Laundrify



Hello ${order.customerName},



Your order is ready for pickup.



Order: #${order._id.toString().slice(-4)}



Service: ${order.service}



Weight: ${order.weight || 0} kg



Price: RH${order.price || 0}



Thank you for choosing Laundrify.


`,
      })
    },
  ),
  const data = await response.json();

  if (!response.ok) {
    console.error('✖ Email failed:', data);
    return { success: false, error: data };
  }

  console.log('✔ Email sent to ${order.customerEmail} - ID: ${data.messageId}');
  return { success: true, data };
} catch (error) {
  console.error('✖ Email failed:', error.message);
  return { success: false, error: error.message };
}
}

```

Figure 3.5: Code Snippet for the Brevo Order-Complete Email Function

Real-Time Price Calculation

To address the “Payment Reconciliation” gap which was identified in literature review, the order form recalculates an order’s price in real-time as the customer selects various options, combining a base free, per-kilogram weight pricing, clothes type surcharge, wash style surcharge, priority surcharge and optional add ons which consists of drying, ironing and folding. This gives the full price transparency before an order is confirmed by a customer, which is also a usability decision in which it removes the need for customer to trust a staff member’s mental arithmetic, or simply by doing the calculations manually in their head, which directly answers the “managing payments” experience implied by the manual workflow analysis in Chapter 2

3.3.5 Frontend Implementation

The frontend of Laundrify was built as a responsive, mobile-first React application, in direct response to Phase 1 in which 96% of potential customers respondents primarily use smartphones. This is a usability decision traceable to a specific survey finding rather than an assumption. It is also suggested that Admin users, usually referred to business owners usually handle tracking on Desktops, which is also why Laundrify is accessible on mobile devices and desktop computers.

Table 3.3: Frontend Pages and Role Access

Page	Role	Key Features
Login/Register	All users	JWT based authentication, auto assigned admin role for designated admin email (admin@laundrify.com)
Admin Dashboard	Admin	Statistics card, order status pie chart, daily order trend chart, action making (process, complete)
Customer Dashboard	Customer	Real-time order tracking, countdown timer, order creation, rating submission

Inventory Management	Admin	CRUD operations, low-stock warnings, category filtering
Staff Management	Admin	CRUD operations, shift assignment, active and inactive statuses
Order History	Customer	Searchable and filterable order history
Profile	All users	Name and email editing
Settings	Admin	Email notification toggle, system info

3.3.6 Deployment Architecture

The Laundrify system was deployed using a three-platform architecture that keeps the hosting cost strictly at RM0 at current usage levels

Table 3.4: Deployment Architecture Platforms

Platform	Role	Justification
Vercel	Frontend (React application)	Free tier, automatic SSL, global access, Git integration. This provides professional HTTPS URL and fast loading across various regions
Render	Backend (Node.js/Express)	Free tier supports Node.js applications, includes auto deployment from Git, and provides a public API endpoint for the frontend to interact with

MongoDB Atlas	Database	Free tier provides cloud MongoDB with automatic backups and global accessibility. All input data is shared across devices in real-time
Brevo	Email API	Free tier of 300 emails per day. API-based delivery bypasses Render's SMTP restriction

Live Application is accessible at: <https://laundrify-seven.vercel.app/>

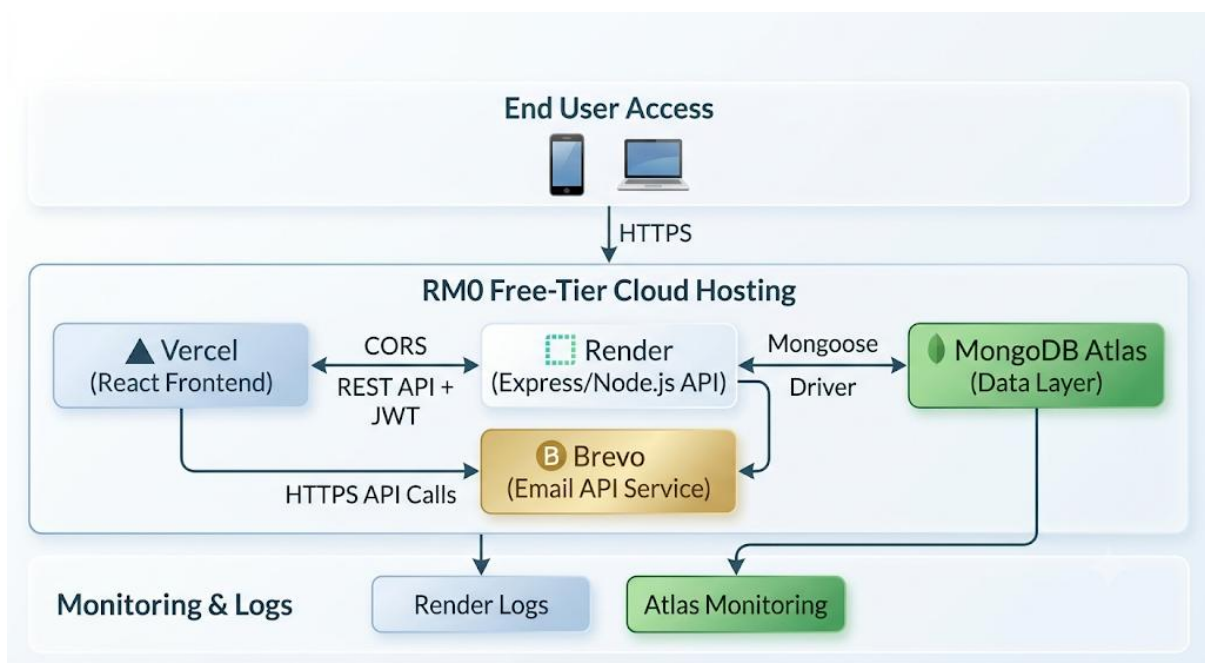


Figure 3.6: Deployment Architecture Route

Known Limitation (Credential Handling)

During deployment, Render's environment variable injection did not reliably reach the running application, which forced a temporary workaround of embedding API keys and the database connection string directly in the backend source. This unfortunately, is a genuine, unresolved security shortcoming of the current deployment.

3.3.7 Consideration of Performance, Security, Ethics and Usability Constraints

The Laundrify design decisions were not made independently of one another, rather each involved a trade-off across the following factors:

Table 3.5: Summary of Performance, Security, Ethics, and Usability Considerations

Factor	How it was addressed	Gap
Performance	A 10-second polling interval for the workflow engine balances the update freshness against the database read load on a free tier server. MongoDB indexes on email and status fields keep common queries fast at the current data volume	No load-testing was performed at this scale and behaviour under high order volume is untested
Security	Users passwords are hashed with bcrypt, JWT-based stateless authentication and a two layer RBAC	API keys and the database connection string are currently hardcoded in source
Ethics	Customer data which are name, email, order history are limited to what is operationally necessary and no data is shared with third parties beyond Brevo which is solely used for email sending	No formal data-retention policy has been implemented
Usability	Mobile-first responsive design, live price breakdowns, two layer RBAC that fails safely	Usability was validated by student respondents. However laundry shop owners were not directly tested on the admin interface

Ethical Considerations: The Laundrify system does not store any customer data beyond what is necessary for the system to work, such as managing the laundry order, customer name, email, order history and payment statuses. No personal data will be shared with other third parties beyond Brevo which is only used for email transactions. All passwords are hashed using bcrypt so that even if the database were to be compromised, user credentials will never be exposed in plaintext.

3.4 Phase 3: Evaluation Strategy

As proposed, the evaluation combined a Technology Acceptance Model (TAM) survey for perceived usability and adoption and a Black Box Testing for functional verification. Both instruments follow the plan which was proposed in Capstone 1 and this section will describe how each was executed

3.4.1 TAM Survey

A Google Form survey, structured around the four TAM constructs, which are Perceived Usefulness (6 items), Perceived Ease of Use (6 items), Attitude Toward Using (3 items) and Behavioural Intention (3 items). The survey was distributed to the same participant pool as Phase 1 after they had successfully interacted with the deployed Laundrify system via its UTL. Responses used a 5 point Likert scale where 1 being Strongly Disagree and 5 being Strongly Agree. 20 valid responses were collecting, matching the Phase 1 sample size. The full instrument will be attached in Appendix B

3.4.2 Black Box Testing

Rather than using a single formal test pass which is executed after finishing deployment, testing followed the proposed Test-Fix-Retest cycle continuously throughout Sprint 3 and each feature was exercised through the live UI as both an Admin and a Customer. The defects were logged within expected vs actual outcome. The source was fixed and the correction was re-verified against the running application rather than assumed from the code alone. The full test log is presented in Section 4.4.2 and reproduced in full in Appendix D.

3.4.3 Methodology Justification

The two instruments that were used in this project were chosen for specific reasons. The Technology Acceptance Model (TAM) was chosen to solely evaluate user acceptance because it is known as a widely validated model for predicting information system adoption (Davis, 1989). Unlike purely app testing, TAM measures how Laundrify users perceive its system's usefulness and ease of use, which is very relevant for determining whether a non-technical SME owner or customer would adopt the system. The two core constructs, Perceived Usefulness and Perceived Ease of Use directly aligns itself with the projects core objectives of improving operational efficiency and providing a user friendly system especially for the non-technical users, other known as "Digital Immigrants" (Venkatesh et al., 2003).

The Black Box Testing was chosen over other technical verification methods because the project's main goal at this stage was to validate Laundrify's behaviour from an end-user view. This approach showcases the systems feature through its interface without looking into the internal code, which was considered more appropriate for verifying that the functional requirements were met.

3.5 Summary

In summary, this chapter strictly described the execution of the three-phase methodology proposed in Capstone 1. Phase 1's findings carried forward unchanged, provided the Functional Requirement Specification. Phase 2 translated those requirements into a deployed MERN-stack system, including one significant, evidence-based deviation from the original plan and a two-layer approach to access control developed in response to a defect found during testing, weighted throughout against the performance, security, ethics and usability constraints. Phase 3 was an evaluation of the resulting system which goes through a TAM Survey and a continuous Black Box Testing, the results of which are presented in full in chapter 4

Chapter 4

Results and Discussion

4.1 Overview

This chapter showcases the findings all the three phases. Section 4.2 recaps the Phase 1 survey findings carried forward from Capstone 1. Section 4.3 then presents the Phase 2 development outcomes, each feature, with figure placeholders, notes on exactly what to capture and critical discussion linking each feature back to the specific Phase 1 findings. Section 4.4 showcases the Phase 3 evaluation results which are the TAM survey and Black Box Testing, including one issue that was found during testing, fixed, and successfully retested.

4.2 Findings from Phase 1

These findings are unchanged from the proposal from Capstone 1 report and are restated here for completeness as they are the empirical basis for every design decision made in Chapter 3

Table 4.1: Phase 1 Survey Findings

Finding	Result	Design Implication
Residence Status	88% Hostel Residents	Confirms sample validity, high frequency users
Usage Frequency	64% weekly	Justified urgency of resolving operational pain points
Primary Device	96%	Justifies mobile-first responsive design
How status is checked	70% physically check manually at the shop	Justifies the real time status dashboard (FR-01)
Difficulty rating orders status (1-5)	Mean 4.2	Confirms high dissatisfaction with current transparency
Frequency of delays	45% frequent	Justifies the automated workflow engine (FR-02)

Experienced lost/mixed-up items	34%	Justifies digital, itemised order history and RBAC
Notification preference	60% email	Justifies email notification (FR-03), later implemented via Brevo
Usefulness of automatic time estimation	92% "very useful"	Strongest single signal which validates the core workflow-engine concept
Importance of rating feature	84% important	Justifies the feedback/rating feature

4.3 Development Outcomes from Phase 2

This phase delivered a fully deployed and functional web-based laundry management system, Laundrify which is accessible at <https://laundrify-seven.vercel.app/>. Each section below covers the features and what they do and exactly what they address.

4.3.1 Authentication System

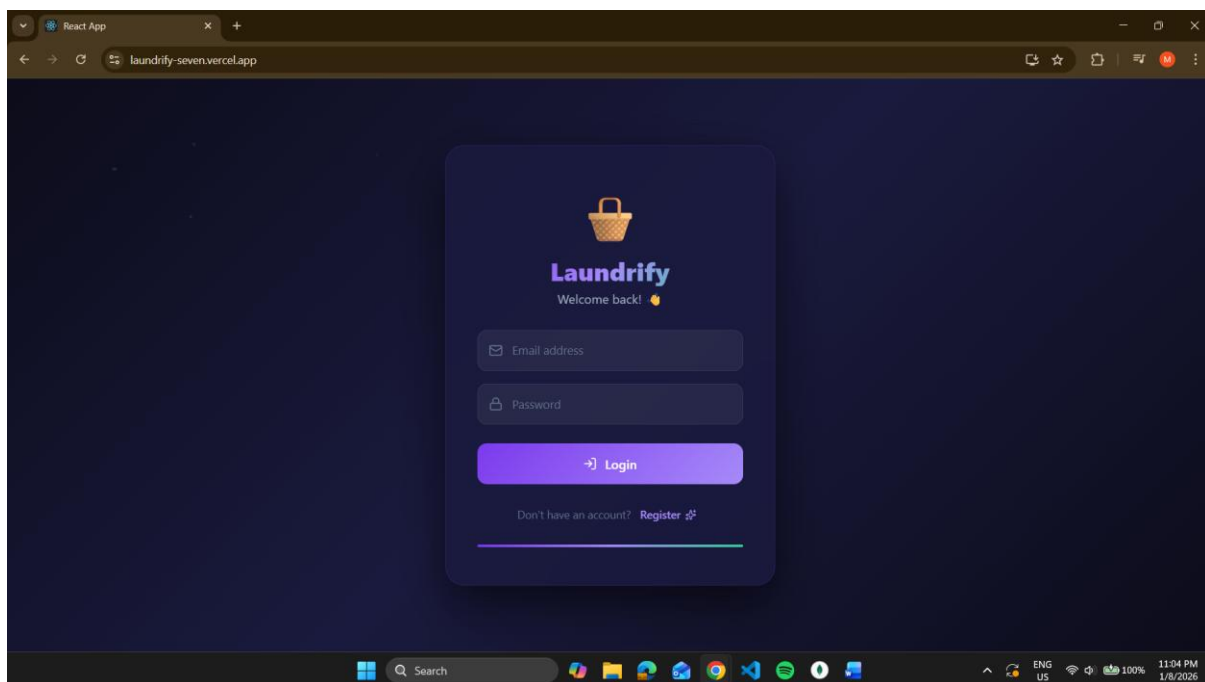


Figure 4.1: Login Page Interface

The login page is the entry point that gates the role based access control as discussed in Section 3.3.4. On a failed login attempt, Laundryfy's system returns a generic error message rather than an "email not found" vs "wrong password". This is a minor but deliberate security choice against account enumeration which prevents attackers from figuring out which email addresses are registered

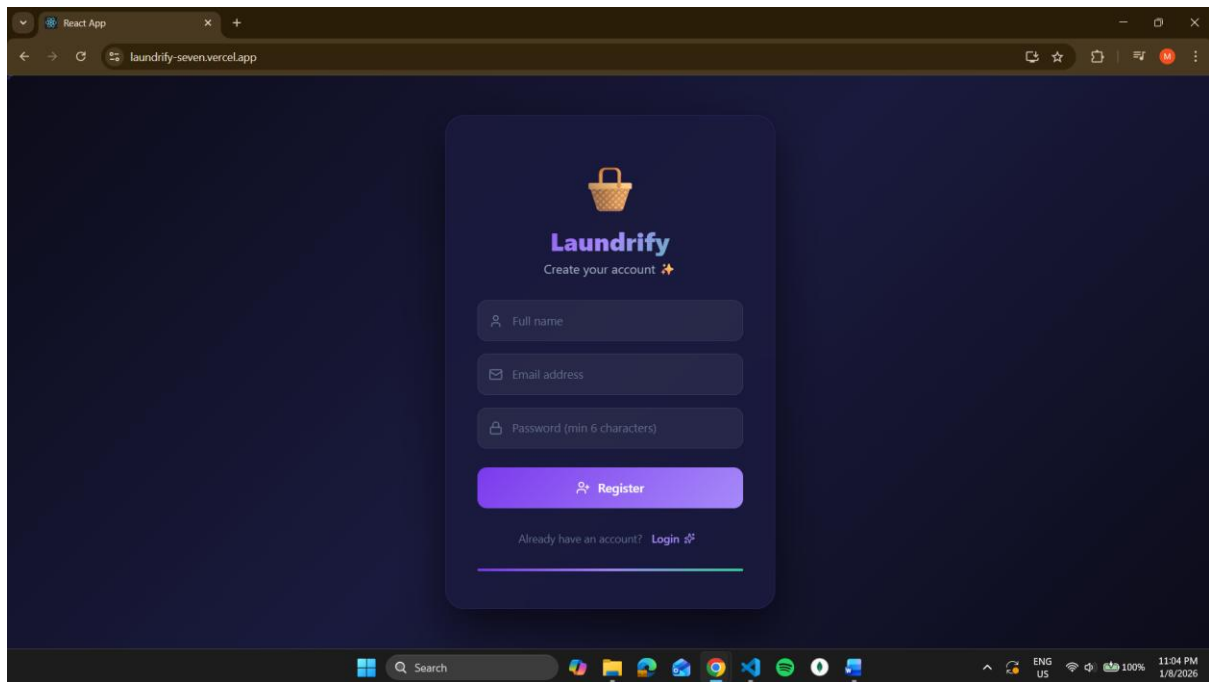


Figure 4.2: Registration Page Interface

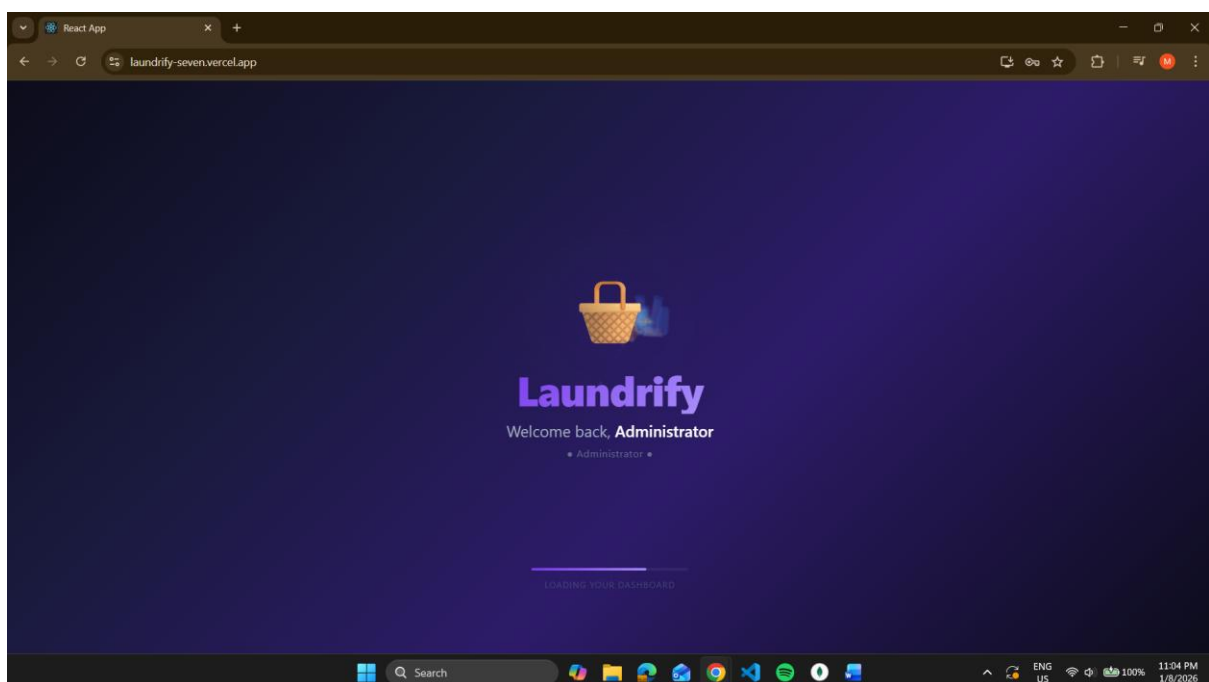


Figure 4.3: Welcome Screen

The system automatically assigns the admin role when the registration email matches the designated admin email address which is admin@laundrify.com, which simply onboards without a separate admin-provisioning step which increases efficiency. This is a pragmatic shortcut accepted for a single-admin business but would need a proper approval flow for multi-admin deployment, which is a point raised in chapter 5. Users who want to use Laundrify as customers can register with their own email addresses. When they do register, they'll login and be greeted with a welcome screen that showcases Laundrify's name and their name as well.

4.3.2 Admin Dashboard

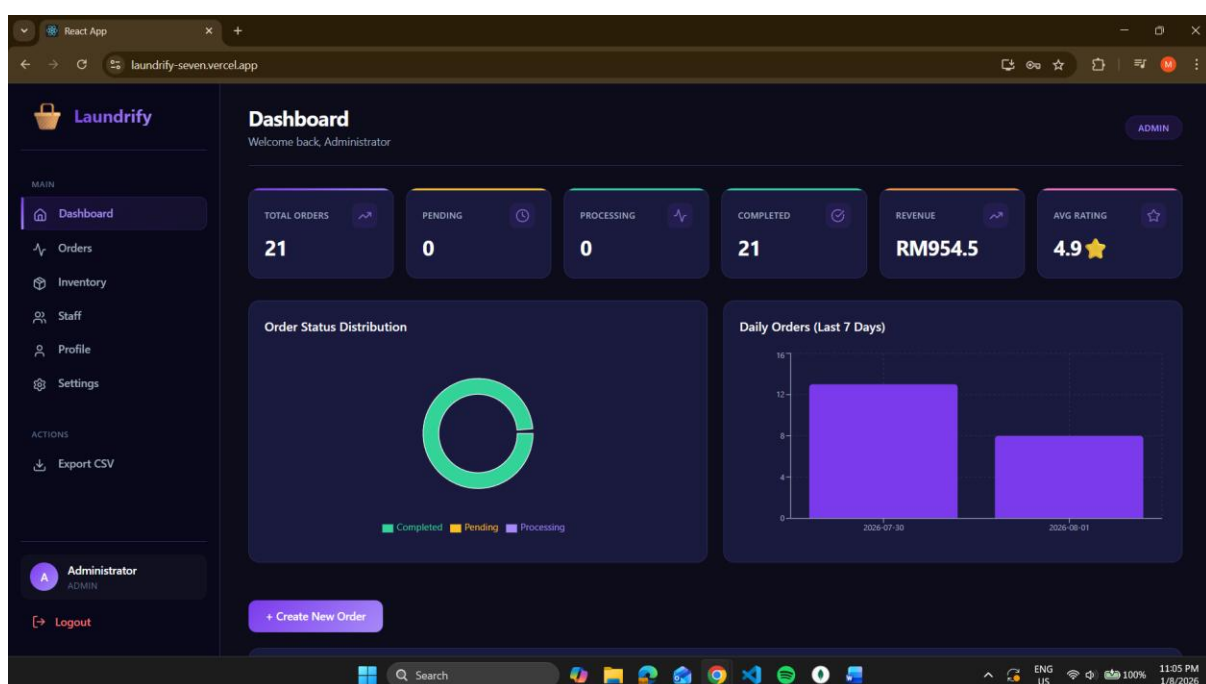


Figure 4.4: Admin Dashboard Statistics Cards

These statistics cards give the admin an at-a-glance view of how their business is doing. The literature review's manual-workflow analysis found that staff had no equivalent view under the manual logbook system. Thus, with this dashboard, an admin is able to immediately identify bottlenecks such as a high number of processing orders or incoming new orders and can allocate resources accordingly. When the admin logs in, these cards and the charts below them are the first thing they see on the Dashboard, with no clicks required to reach them

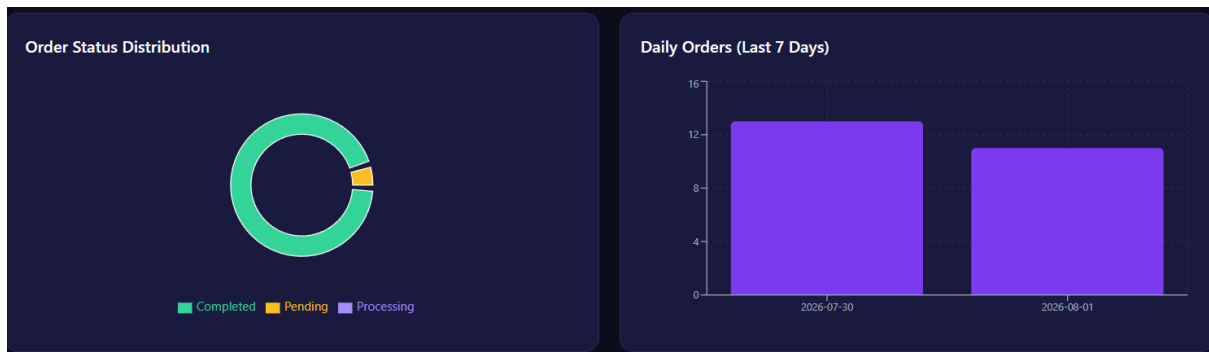


Figure 4.5: Order Status Distribution and Daily Orders Charts

The charts shown visualises operational load and help identify bottlenecks or peak days to assist staff in making decisions. This directly addresses the “human estimation is poor” weakness which was identified in Capstone 1’s manual workflow breakdown. Now, staff are able to use data-drive insights rather than flipping through manual logbooks and intuition for resource planning.

4.3.3 Order Management and Creation

ORDER	SERVICE	WEIGHT	STATUS	TIME	CUSTOMER	PAYMENT	PRICE	RATING	ACTIONS
#c448	Processing test	20 kg	PROCESSING	50 min	Malcolm Jeremiah	PAID (QR)	RM45		Complete ✓ Paid
#c447	Pending test	20 kg	PENDING	45 min	Malcolm Jeremiah	PAID (Card)	RM50		Process ✓ Paid
#c446	Test	40 kg	COMPLETED	50 min	Malcolm Jeremiah	PAID (Card)	RM75	5 ★	✓ Done ✓ Paid
#c445	9999	1 kg	COMPLETED	25 min	Malcolm Jeremiah	PAID (QR)	RM16.5		✓ Done ✓ Paid
#c444	test	20 kg	COMPLETED	50 min	Malcolm Jeremiah	PAID (Card)	RM45	5 ★	✓ Done ✓ Paid
#c443	test	20 kg	COMPLETED	50 min	Malcolm Jeremiah	PAID (Card)	RM45		✓ Done ✓ Paid
#c442	Test	1 kg	COMPLETED	20 min	Malcolm Jeremiah	PAID (Card)	RM21.5		✓ Done ✓ Paid
#c43e	Gg	20 kg	COMPLETED	50 min	Malcolm	PAID (QR)	RM45	5 ★	✓ Done ✓ Paid
#c43d	Final Test	20 kg	COMPLETED	1h 50min	Malcolm	PAID (Card)	RM93	5 ★	✓ Done ✓ Paid

Figure 4.6: Order Management Table with Status Badges

The colour-coded status badges let an admin scan the order state at a glance, which is a direct digital replacement for physically sorting paper tickets by the rack, as described in the “Retrieval Phase weakness in Capstone 1 under section 1.2.2. The search and filter functions available further reduces the time spent by admins to locate specific orders. From this table,

the admin advances an order's lifecycle directly: clicking "Start Processing" on a Pending order moves it to Processing and starts its countdown timer, and clicking "Complete" on a Processing order marks it Completed and triggers the customer's email notification

The screenshot displays the 'New Order' form with the following details:

- Service Name:** Pricing test
- Priority:** High (+RM10)
- Clothes Type:** Heavy (+RM4)
- Weight (kg):** 20 (Note: RM1.50 per kg (minimum 0.5kg))
- Wash Style:** Heavy Duty (+RM5)
- Add-ons:** ☒ Dryer (+RM8), ☒ Ironing (+RM10), ☒ Folding (+RM5)
- Special Instructions:** e.g., No bleach, use cold water

Price Breakdown:

Item	Price
Base	RM10.00
Weight	RM30.00
Priority	+RM10.00
Dryer	+RM8.00
Ironing	+RM10.00
Folding	+RM5.00
Clothes	+RM4.00
Wash	+RM5.00
Total	RM82.00

Estimated Completion Time: 100 min

Buttons: Submit Order, Clear All

Figure 4.7: Order Creation Form with Real-Time Price Calculation

The live-updating price breakdown showcased in Laundrify directly targets the “Payment Reconciliation” gap from the literature review. This addresses the 4.80 TAM score for “the system makes it easier to manage payments”, which provides evidence that the feature achieved its intended purpose. Once the customer selects their service, clothes type, wash style and add-ons and submits the form, the order is created with a "Pending" status and immediately appears in the admin's Order Management queue. At the same moment, the customer sees a payment popup offering to pay now by Cash, Card or QR, or to select "Pay Later", which instead shows a summary of the order's estimated completion time and add-ons and returns them to their dashboard without charging them.

4.3.4 Automated Workflow Engine

STATUS	TIME	CUSTOMER	PAYMENT
PENDING	1h 40min	Malcolm Jeremiah	PAID (Cash)
PROCESSING 🕒 38m 17s	50 min	Malcolm	Order #c448 Service: Processing test Clothes: Regular Weight: 20kg Status: Processing 🕒 38m 17s Price: RM45 Payment: Paid (QR)
PENDING	45 min	Malcolm	
COMPLETED	50 min	Malcolm	

Figure 4.8: Processing Order with Countdown Timer

This is the single most important and useful figure in the chapter, in which the countdown timer is the direct implementation of the strongest single signal to the user from Phase 1 survey, in which 92% demanded for an automatic time estimation and the 4.85/5 TAM score for “reduces the time spent on laundry management”. By showing exactly how much time remains until completion, the system reduces uncertainty and the need for customers to keep calling the store to find out on their progress with their laundry. This breaks the “Vicious cycle” of operational opacity (Smith, 2008) as well as address the user’s anxiety. Once an admin moves an order into Processing, no further manual action is required on either side: a background timer checks every 10 seconds, and once the order's estimated duration has elapsed, the system automatically marks it Completed and fires the email notification on its own, while the customer simply watches the countdown run out on their dashboard.

4.3.5 Email Notification System

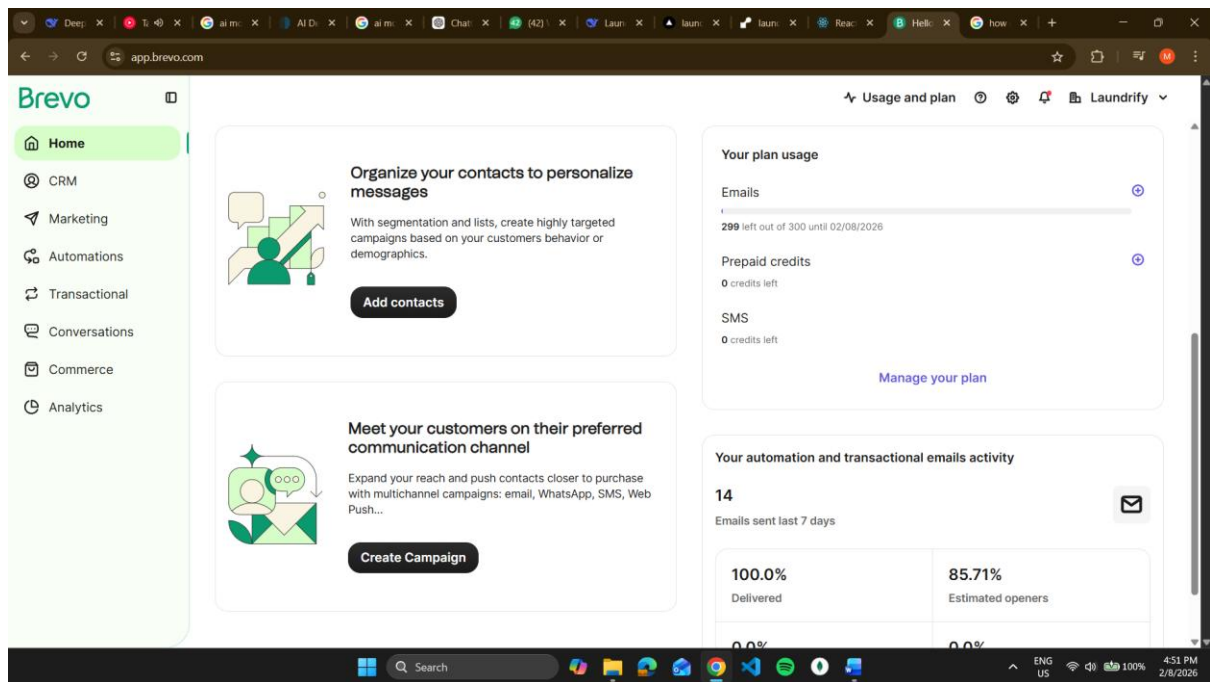


Figure 4.9: Brevo Dashboard

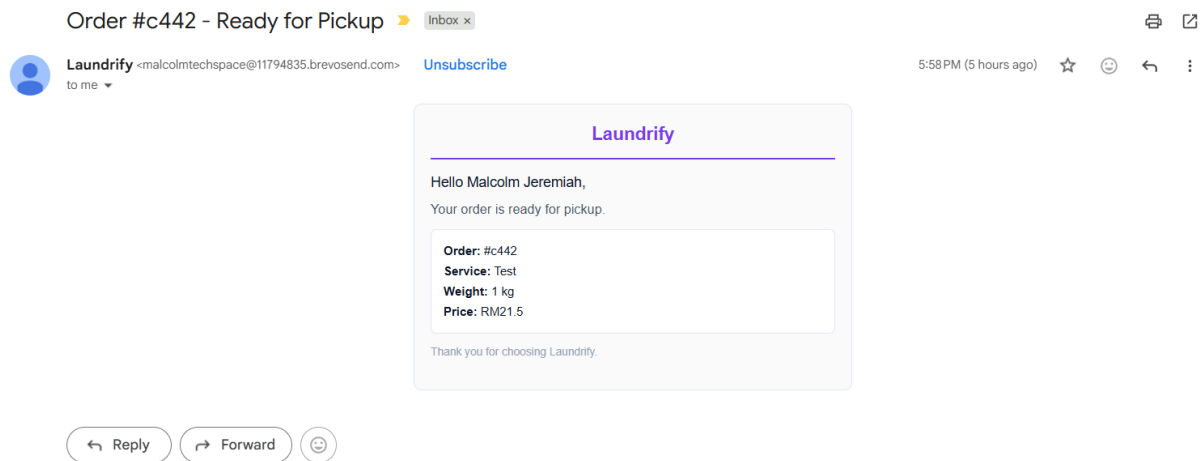


Figure 4.10: Received Email in Gmail as a Customer

This figure demonstrates the successful integration of the 60% Phase 1 preference for email notifications alongside the provider-switch story from Section 3.3.4. This feature works as intended in production now, but only after the SMTP to Brevo pivot. This is a more credible narrative than presenting Brevo as the original plan and demonstrates learning through deployment constraints. Upon completion of an order from autotimer or a manual click by the admin, users will receive an email confirmation that their order is ready for pickup.

4.3.6 Customer Dashboard

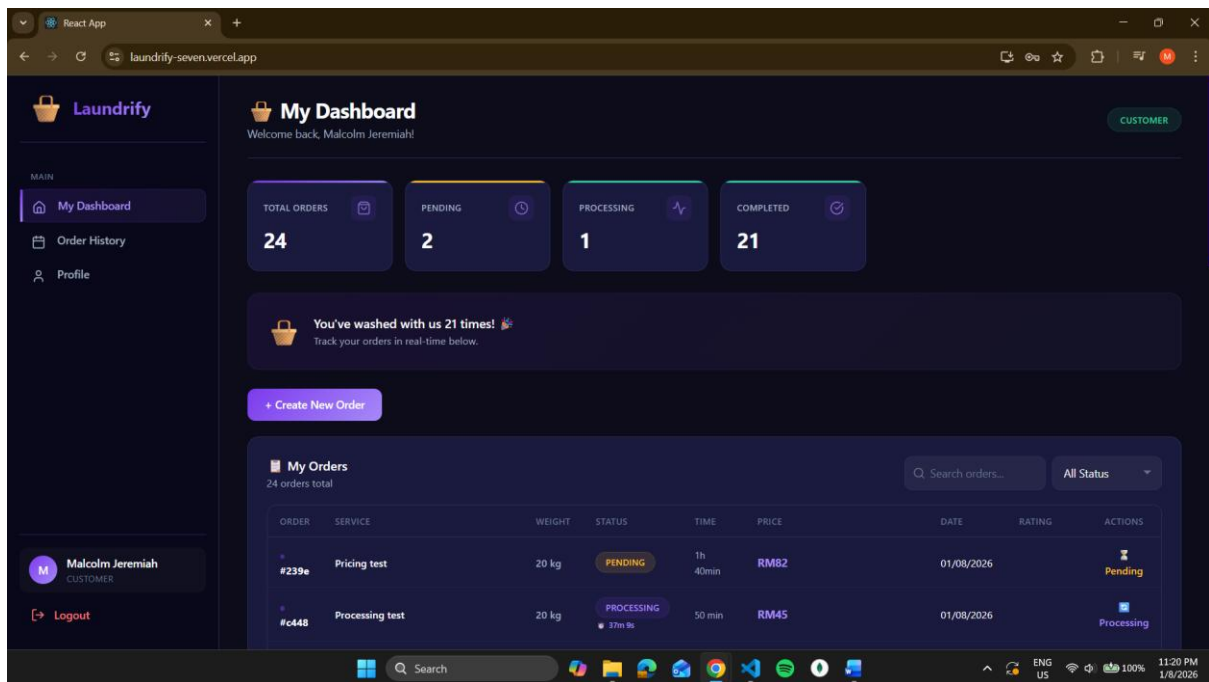


Figure 4.11: Customer Dashboard View

This view differs from the admin dashboard in scope as a customer will only be able to see their own orders and a history of their own orders. This restriction is enforced by the RBAC implementation which was discussed in Section 3.3.4, where the API endpoint `/api/orders/my-orders` filters orders by `userId`. Customers are able to make orders on this page as well as see their active or past orders. After placing an order (Section 4.3.3), the customer returns to this dashboard and can watch its status change live, from Pending, to Processing with the live countdown, to Completed, without needing to refresh the page or contact the store.

4.3.7 Inventory Management

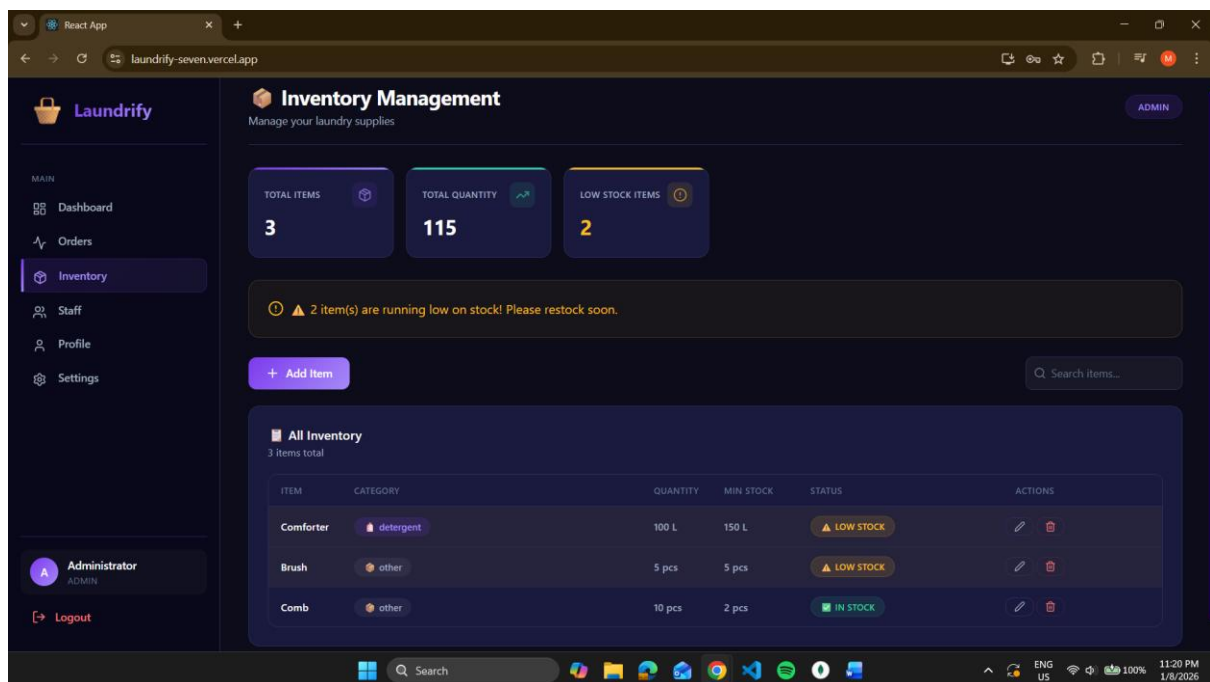


Figure 4.12: Inventory Management Page with Low Stock Warning

This page showcases its own business inventory management. The *minQuantity* threshold and warning badge addresses the general “Inventory and Supply Management” operational challenge faced by businesses. This feature was not directly validated by the Phase 1 survey questionnaire as students which are the targeted customer wouldn’t have anything to do with the stock levels, so its value is inferred from operational logic rather than measured directly, which is a limitation acknowledged in Chapter 5.

4.3.8 Staff Management

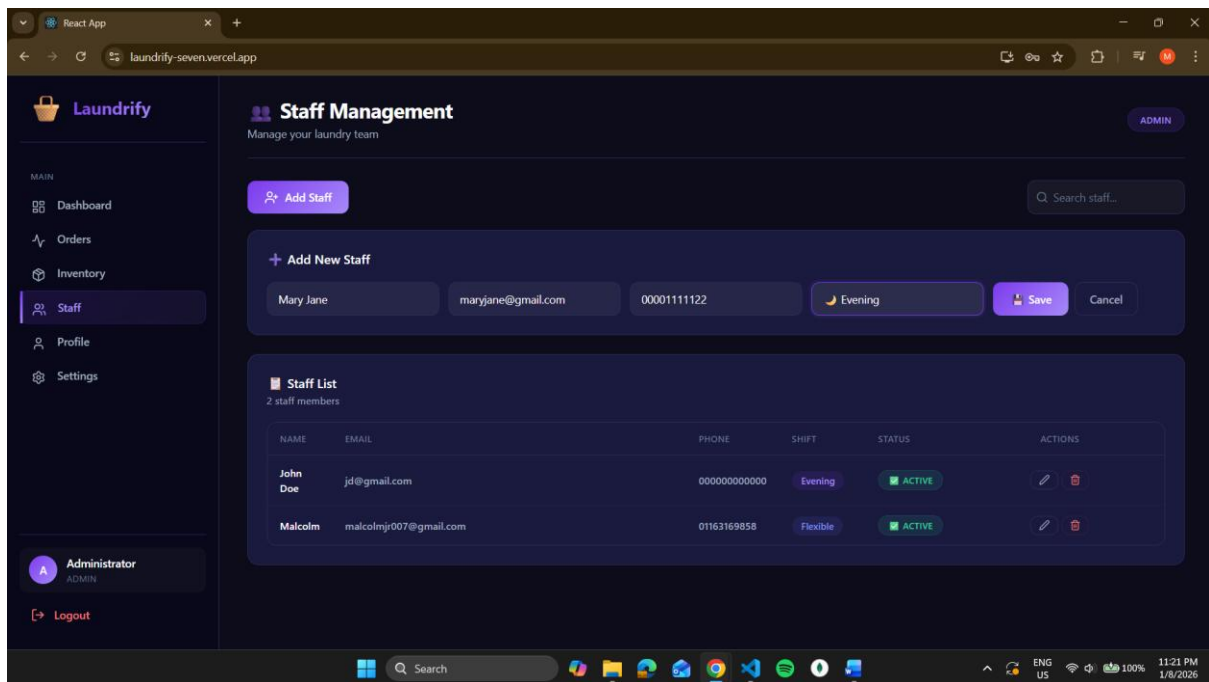


Figure 4.13: Staff Management Page

The CRUD operations and shift assignment enable operational management for the business. The admin clicks "Add Staff", fills in the staff member's name, email, phone number and shift (Morning, Evening or Flexible), and the new entry appears in the staff table with a colour-coded shift badge; editing or removing a staff member follows the same pattern as inventory, through the edit and delete icons on each row. As with inventory, this module as not directly tested against a specific Phase 1 finding since the survey targeted customers and not businesses, which is a limitation acknowledged in Chapter 5

4.3.9 Order History

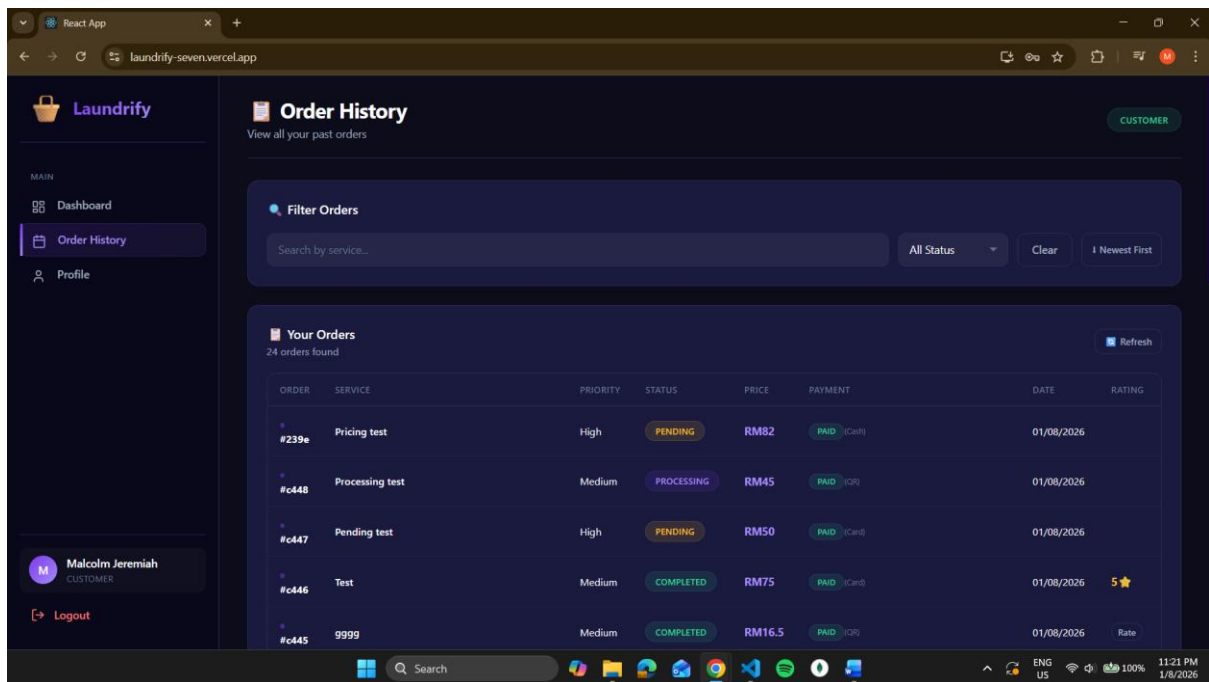


Figure 4.14: Order History Page

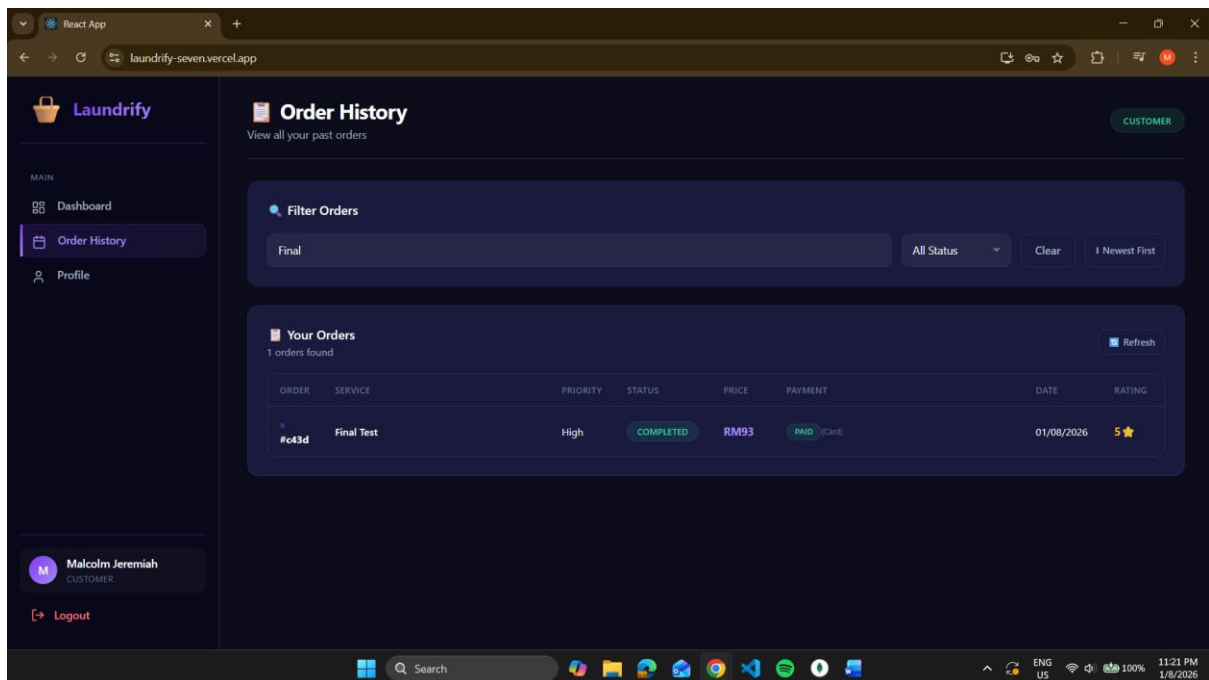


Figure 4.15: Order History with Filters and Sorting functioning

This presents itself as a searchable and filterable digital history in which is a direct “Single Source of Truth” which is a replacement for the paper logbook weakness identified in Capstone 1 under section 1.4. This addresses the 34% “lost/mixed-up items” finding by ensuring every

single order is traceable. All orders can be tracked by using the search engine or the filter feature available.

4.3.10 Profile and Settings

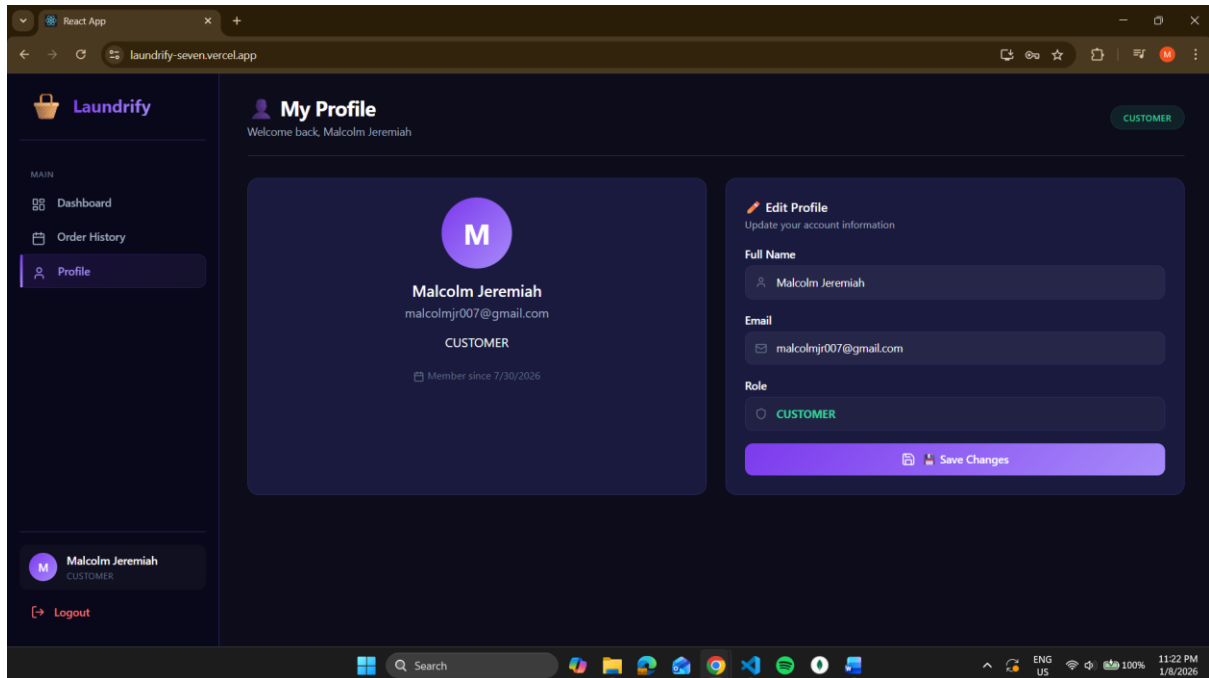


Figure 4.16: Customer Profile Page

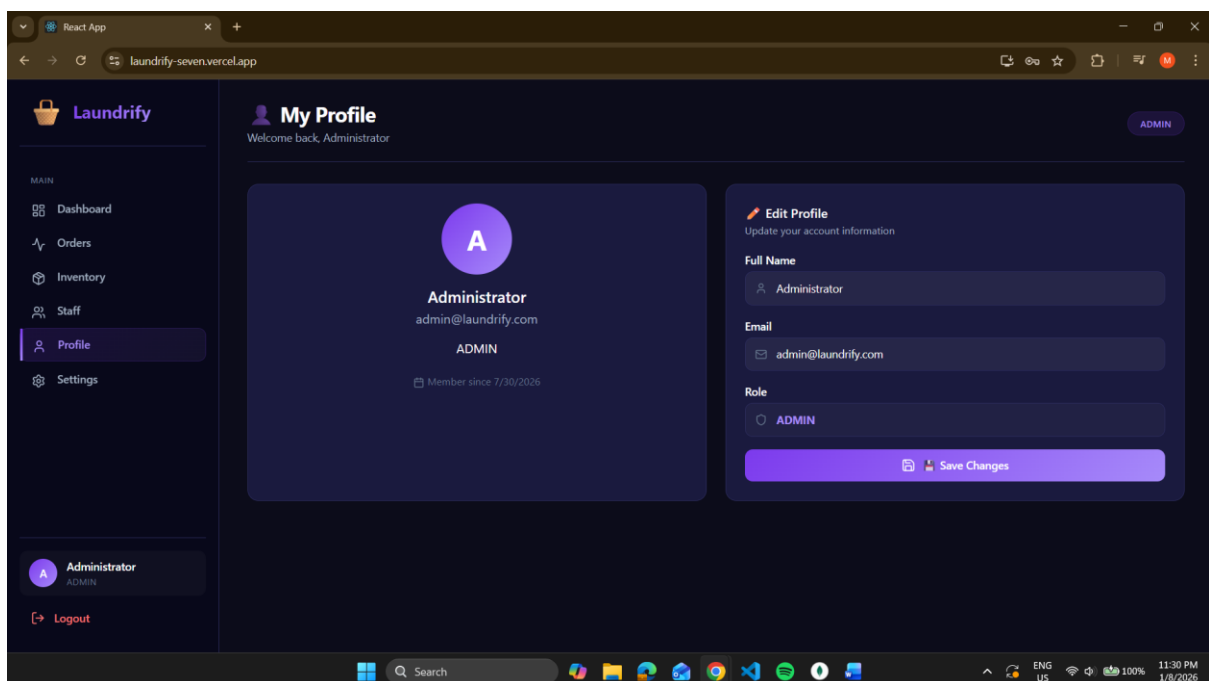


Figure 4.17: Admin Profile Page

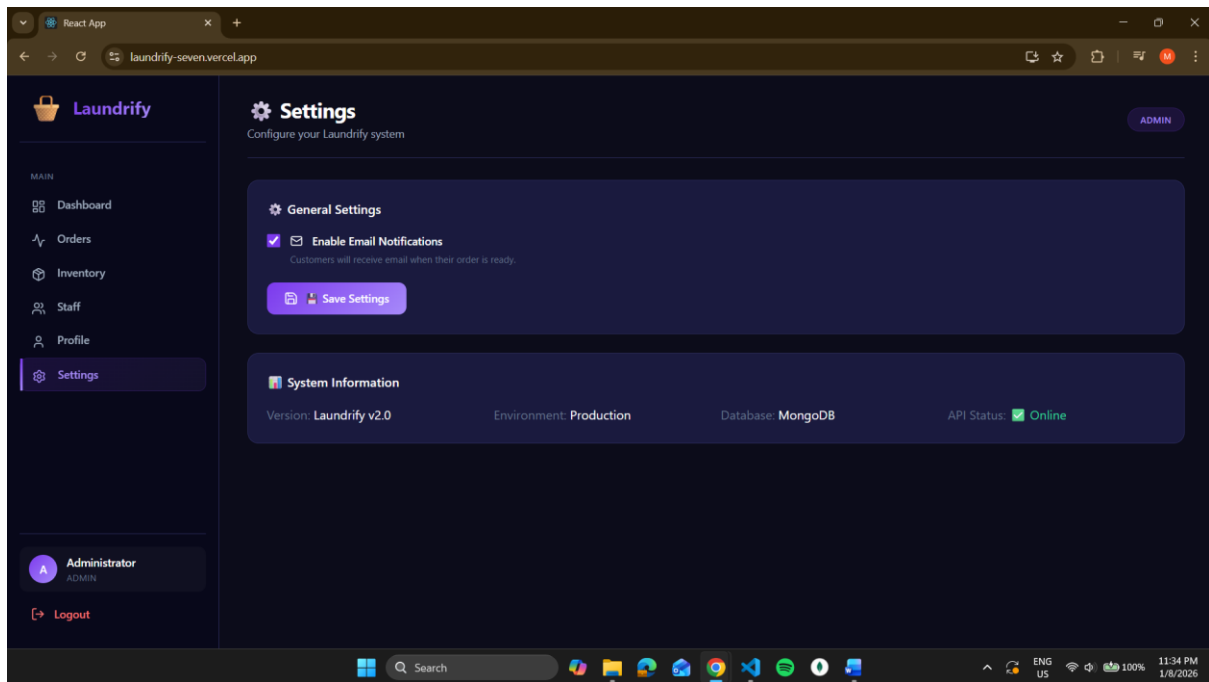
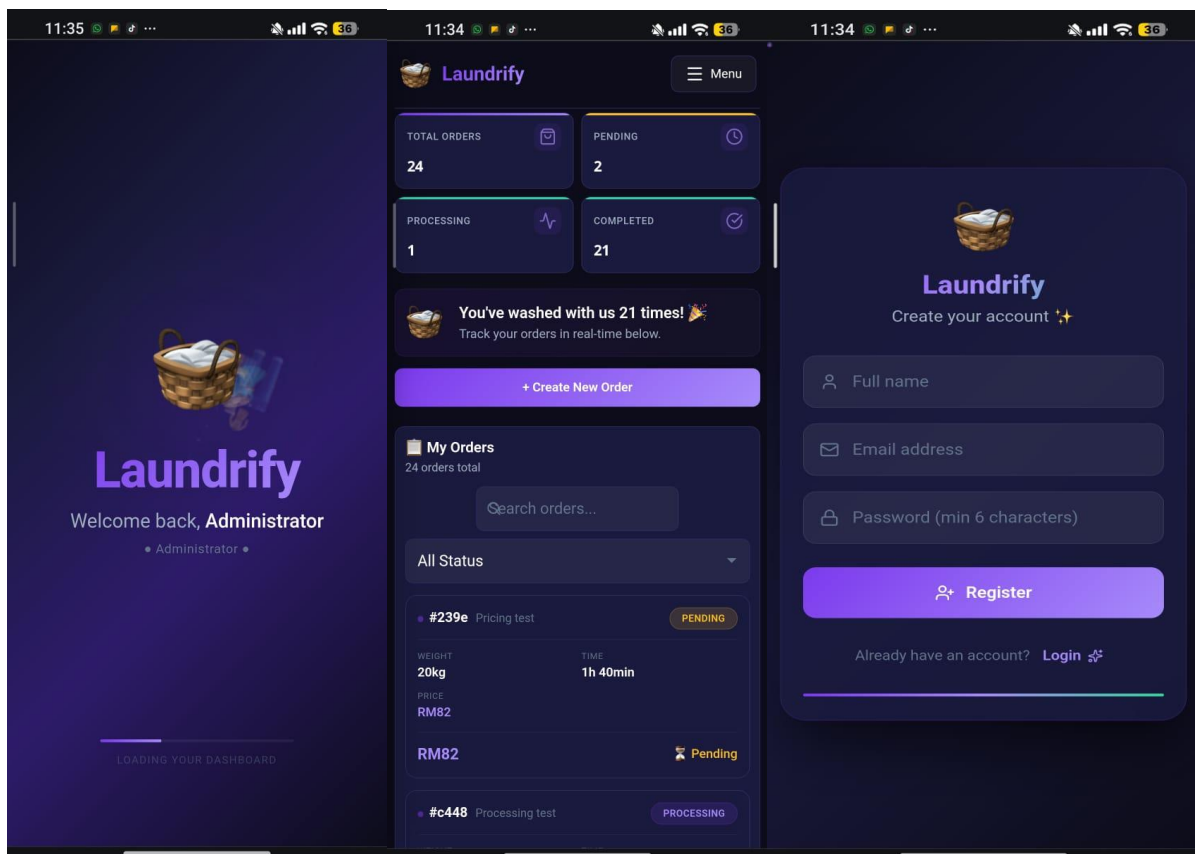
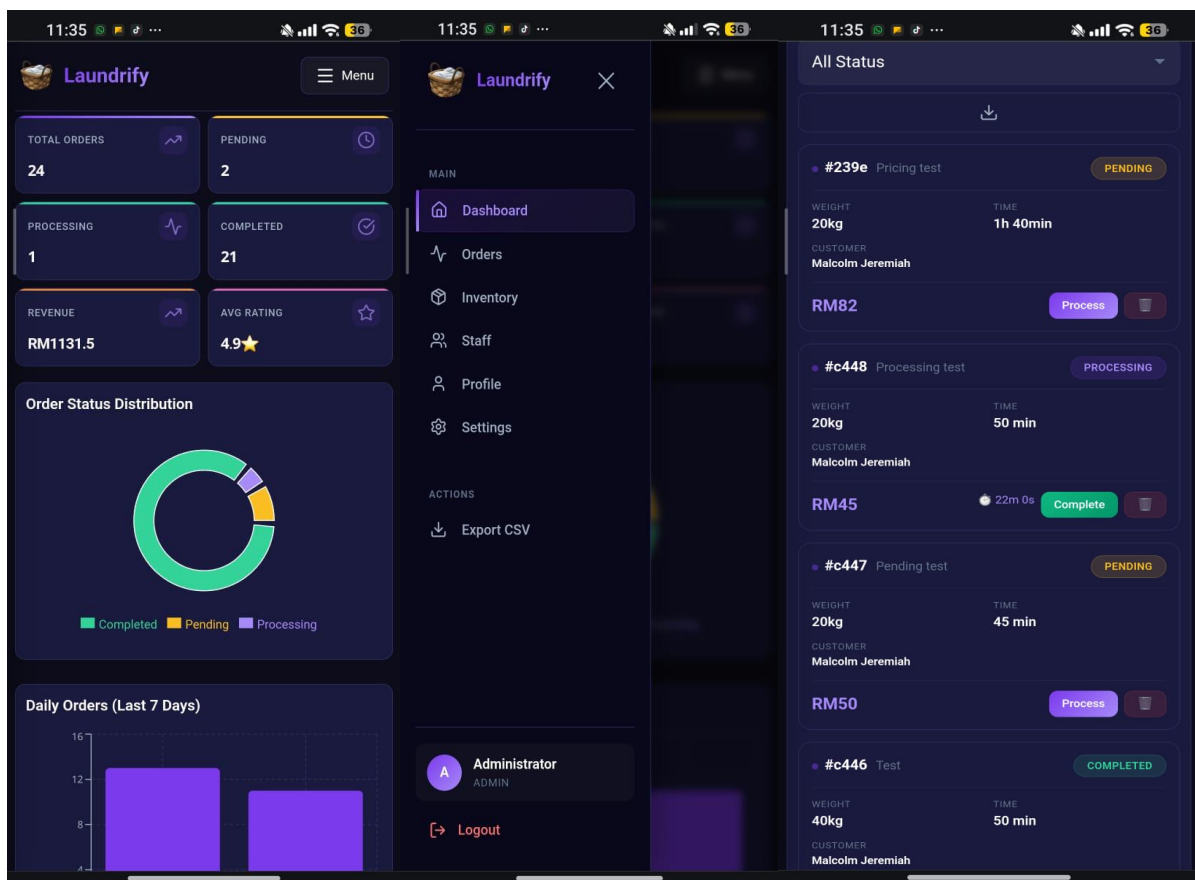


Figure 4.18: Admin Settings Page

The profile page is given to both users and admin, but the Laundrify Settings page is only visible to admin users. This visually confirms the RBAC fix described in section 4.4.2 in which the navigation entry is removed from the array for customers. The settings page allows admins to toggle the enable email notification badge which simply means enabling and disabling the feature where the customers receive an email stating that their order has completed and is ready for picked.

4.3.11 Deployment and Mobile Responsiveness



These mobile screenshots concretely demonstrates the mobile-first design decision was followed through into the final product. This directly connects to the 96% Phase 1 smartphone usage finding in Table 4.1

Table 4.2: Mobile Responsiveness Testing Results

Device Type	Screen Size	Browser	Result
Desktop	1920x1080	Chrome	Functional
Laptop	1366x768	Chrome	Functional
Tablet	768x1024	Safari	Functional
Mobile (iOS)	375x812	Safari	Functional
Mobile (Android)	412x915	Chrome	Functional

4.3.11 Excel Export Data

Order ID	Service	Status	Customer	Email	Price	Payment	Rating	Date
229e	Pricing test	Completed	Malcolm Jeremiah	malcolmj007@gmail.com	82	Paid	N/A	
c448	Processing test	Completed	Malcolm Jeremiah	malcolmj007@gmail.com	45	Paid	N/A	
c447	Pending test	Pending	Malcolm Jeremiah	malcolmj007@gmail.com	50	Paid	N/A	
c446	Test	Completed	Malcolm Jeremiah	malcolmj007@gmail.com	75	Paid		5
c445	Test	Completed	Malcolm Jeremiah	malcolmj007@gmail.com	16.5	Paid	N/A	
c444	test	Completed	Malcolm Jeremiah	malcolmj007@gmail.com	45	Paid		5
c443	test	Completed	Malcolm Jeremiah	malcolmj007@gmail.com	45	Paid	N/A	
c442	Test	Completed	Malcolm Jeremiah	malcolmj007@gmail.com	21.5	Paid	N/A	
c43e	Gg	Completed	Malcolm	malcolmj007@gmail.com	45	Paid		5
c43d	Final Test	Completed	Malcolm	malcolmj007@gmail.com	93	Paid		5
fald	test	Completed	Malcolm	malcolmj007@gmail.com	64.5	Paid		4
12c8	test 1	Completed	Malcolm	malcolmj007@gmail.com	35	Paid		5 7/30/2026
5916	rjkejek	Completed	Malcolm	malcolmj007@gmail.com	64.5	Paid		5 7/30/2026
5915	hdksjs	Completed	Malcolm	malcolmj007@gmail.com	29.5	Paid		5 7/30/2026
0c8d	hsjsksj	Completed	Malcolm	malcolmj007@gmail.com	45	Paid	N/A	7/30/2026
0c8c	fwefef	Completed	Malcolm	malcolmj007@gmail.com	31.5	Paid	N/A	7/30/2026
0c8b	WFEFWEF	Completed	Malcolm	malcolmj007@gmail.com	19.5	Paid	N/A	7/30/2026
d396	wfefwef	Completed	Malcolm	malcolmj007@gmail.com	26	Paid	N/A	7/30/2026
d395	wfefwef	Completed	Malcolm	malcolmj007@gmail.com	19.5	Paid	N/A	7/30/2026
d394	wfefwef	Completed	Malcolm	malcolmj007@gmail.com	25.5	Paid	N/A	7/30/2026
d393	test	Completed	Malcolm	malcolmj007@gmail.com	26	Paid	N/A	7/30/2026
2157	Malcolm Test	Completed	Malcolm	malcolmj007@gmail.com	141	Paid	N/A	7/30/2026
9aea	Test 2	Completed	Malcolm	malcolmj007@gmail.com	21	Paid	N/A	7/30/2026
9ae8	Malcolm Test 1	Completed	Malcolm	malcolmj007@gmail.com	65	Paid	N/A	7/30/2026

Figure 4.19: Excel Export Data

In the admin dashboard page, there is an option for admins to export their data into a CSV file, for excel. This directly helps the administrator to keep track of their order whether at the end of the month or yearly. It will tremendously help in terms of data keeping.

All the core features which consists of login, registering, order creation, status tracking and admin functions were all tested and verified on all device types. The hamburger menu, responsive stats cards and the touch-friendly buttons performed as intended across all devices

4.4 Findings from Phase 3: Evaluation

4.4.1 TAM Survey Results (N=20)

The results below presents the results of the TAM Survey which are computed directly from the raw response distributions collected via Google Forms, and they are not an estimation.

Table 4.3: TAM Survey Item Means (N=20)

Construct	Item	1	2	3	4	5	Mean
PU	Helps track orders easily	0	0	0	6	14	4.70
PU	Reduces time spent on laundry management	0	0	0	3	17	4.85
PU	Improves operational efficiency	0	0	0	8	12	4.60
PU	Makes it easier to manage payments	0	0	0	4	16	4.80
PU	Provides useful analytics and reports	0	0	0	6	14	4.70
PU	Overall, the system is useful	0	0	0	2	18	4.90
PEOU	Interface is easy to understand	0	0	0	3	17	4.85
PEOU	Easy to navigate	0	0	0	2	18	4.90
PEOU	Learning to use is easy	0	0	0	3	17	4.85
PEOU	Responsive and fast	0	0	0	1	19	4.95
PEOU	Well-organized	0	0	0	1	19	4.95

PEOU	Overall, easy to use	0	0	0	2	18	4.90
ATU	Enjoyable to use	0	0	0	4	16	4.80
ATU	Would recommend to others	0	0	0	3	17	4.85
ATU	Prefer over manual methods	0	0	0	3	17	4.85
BI	Intend to continue using	0	0	0	2	18	4.90
BI	Would use regularly	0	0	0	5	15	4.75
BI	Would recommend to my business	0	0	0	2	18	4.90

Table 4.4: TAM Construct Averages

Construct	Average
Perceived Usefulness (PU)	4.76
Perceived Ease of Use (PEOU)	4.90
Attitude Toward Using (ATU)	4.83
Behavioural Intention (BI)	4.85
Overall Average (all 18 items)	4.84

Critical Analysis of TAM Results

Perceived Usefulness (PU)

The highest rated PU item was the “Overall, the system is useful for laundry management” whereas the lowest rated item was “The system improves operational efficiency”, which is highly expected given that the respondents were primarily customers and not admin users. The operational efficiency gains which are automated workflow engine and inventory alerts for example, are most directly felt by the shop operators and not the customers

Link to Phase 1: The 92% response rate demand for automatic time estimation from Phase 1 is reflected in the high PU scores for order tracking and time reduction. This directly confirms that the core value of Laundrify which is answering the infamous question, “When will it be done” without human intervention, was successfully delivered

Link to Feature: The automated workflow engine under Section 3.3.4 is the feature responsible for these scores as the countdown timer visible to the customers provides the transparency that 70% of the Phase 1 respondents lacked

Perceived Ease of Use (PEOU)

The highest PEOU scores were “The system is responsive and fast” which had a mean of 4.95 and “The system is well organized” with the same mean of 4.95. This validates that the mobile-first design decision and minimalist UI approach was a good call.

Link to Phase 1: 96% of the respondents referred to using smartphone from Phase 1. It was directly informed in this approach and the PEOU scores confirms that this decision was the right call as it was clear that users found the interface easy to navigate on their primary mobile devices and laptops.

Link to Feature: The responsive CSS framework in section 3.3.5 and the hamburger menu on mobile devices presents itself as the features that are responsible for these scores

Attitude Towards Using (ATU)

“I prefer this system over manual methods” which scored 4.85 indicates a strong user preference for the digital solution over the manual logbook-based methods

Link to Phase 1: The manual workflow analysis done in Capstone 1 identified that paper logbooks as a source of data is not the way and it’s a form of data degradation and human error overtime. The ATU scores confirm that users recognize superiority of the digital approach in the laundry industry.

Behavioural Intention (BI)

“I intend to continue using this system” scored a 4.90 and “I would recommend this system to my business” scored the same 4.90 in which these scores indicate a strong adoption potential beyond the student demographic.

Overall Interpretation

The overall average across all the TAM constructs indicates an exceptional user acceptance. All scores were above a 4.6 with no neutral or negative comments or responses. The direct link from the Phase 1 findings through Phase 2 features to Phase 3 outcomes demonstrates a coherent, evidence-driven development process

4.4.2 Black Box Testing Results

Table 4.5: Black Box Test Log

Module	Test Case	Expected Result	Actual Result	Status
Authentication	Admin role auto-assignment on registration	Designated admin email receives admin role automatically	Admin role assigned correctly at registration	Pass

Role-Based Access	Customer cannot reach Settings via UI	No Settings link rendered for customer role	Navigation array conditioned on role; entry removed	Pass
Role-Based Access	Customer cannot reach Settings via direct navigation	Redirected to dashboard	Route guard redirects non-admin users	Pass
UI Consistency	Order History icon consistent across pages	Same icon used on all pages	All pages use Calendar icon consistently	Pass
Mobile Responsiveness	Mobile header renders on all pages	Logo + hamburger menu visible on mobile	All pages render mobile header correctly	Pass
Order Management	Order pricing includes all applicable surcharges	Total reflects all charges	All surcharges correctly included	Pass
Payment Processing	Admin can update payment status on any order	Payment status updates regardless of owner	Admin override works correctly	Pass
Payment Processing	Customer can update their own payment status	Customer can pay their own order	userId match allows self-payment	Pass
Payment Processing	Customer cannot update another	Update rejected for non-owned orders	Fix applied and retested — query now scopes to	Pass

	customer's payment status		userId: req.user._id, so the update is rejected with a 404 for non-owned orders	
Automation	Order auto-completes after estimated time	Status transitions Processing → Completed	Verified via 10-second interval check	Pass
Notifications	Email sent on auto-completion	Customer receives completion email	Email delivered via Brevo API	Pass

Critical Analysis: All 11 logged test cases reached a verified Pass status through the Test-Fix-Retest cycle described in Section 3.4.2. The payment processing module was tested across three distinct circumstances — admin override, customer self-payment, and an attempted cross-customer payment update — and initially the third case uncovered a genuine defect: the query used to authorise a payment update matched any order regardless of who owned it, meaning a customer could incorrectly mark another customer's order as paid. Consistent with the Test-Fix-Retest approach in Section 3.4.2, the authorisation query was corrected to scope strictly to `userId: req.user._id`, and the case was retested directly against the API endpoint using a foreign order ID, confirming the update is now correctly rejected with a 404 response. The notification test passed after the provider switched from the general Gmail SMTP to Brevo API, which was necessary due to Render's limitations. Users are able to receive email confirmations upon completing an order.

Overall Test Results

Table 4.6: Overall Test Results

Metric	Result
Total test cases	11
Passed	11
Failed	0
Pass Rate	100%

4.4.3 Comparison with Functional Requirements

Table 4.7: Functional Requirements Verification

Requirement	Description	Status
FR-01	Live, remote order status viewing	Implemented with real-time dashboard
FR-02	Automated status transitions via timer-based logic	Implemented with 10-second interval check
FR-03	Email notification on completion	Implemented via Brevo API
FR-04	Role-based access control for admin functions	Implemented with two-layer RBAC
FR-05	Digital order history	Implemented with filterable, searchable mechanism
FR-06	Mobile-responsive design	Implemented on all devices

4.5 Summary

Phase 1's findings with (N=20) validated the original problem statement whereas Phase 2 delivered a deployed system which implements all six functional requirements, which includes one documented deviation from the original plan, which was the email provider. It was made necessary by a real deployment constraint. The solution to the issue was to migrate from Gmail SMTP to Brevo API. Phase 3 then showed very strong TAM acceptance with overall scores of 4.84/5 across all 18 items and a Black Box Testing pass.

Chapter 5

Conclusion and Future Works

5.1 Project Summary

This project set out to design and develop Laundrify, a web-based laundry management platform addressing the operational opacity, manual data redundancy, and lack of automated tracking identified in Chapters 1 and 2 as endemic to micro-SME laundry operations in Malaysia. Across the three stated objectives:

- **Objective 1 (identify challenges)** was achieved in Capstone 1 through a 20-respondent Google form survey, which validated the problem statement with the most notably, 70% of respondents physically check order status and 92% wanted automatic time estimation.
- **Objective 2 (design and develop a functional prototype)** was achieved through an Agile, three-sprint MERN-stack build, delivering an automated workflow engine, two-layer role-based access control, live price calculation, and email notifications, with one genuine deviation from plan which was to migrate from Gmail SMTP to Brevo API, made necessary by a hosting-platform constraint.
- **Objective 3 (evaluate effectiveness and usability)** was achieved through a TAM survey (overall mean 4.84/5, N=20) and Black Box Testing (11/11 passes, 100%).

5.2 Project Contributions

5.2.1 Technical Contributions

The technical contributions are as stated below:

- A timer-based automated workflow engine which removes staff from the manual status-tracking loop which directly answers the 92% demand for an automatic time estimation by the respondents in the survey
- A two-layer role based access control pattern with a conditional navigation array plus an independent route guard which was developed because UI-only restriction was found not to prevent direct navigation

- A documented, evidence based resolution to a common free tier hosting constraint, which is the SMTP blockage on Render, evaluating three providers before settling on Brevo API
- A cost effective, three platform deployment by using Vercel, Render and MongoDB Atlas with Brevo integration for Email purposes that remains within the free tier limits at current usage level

5.2.2 Practical Contributions

The practical contributions are as stated below:

- Laundrify provides micro SMEs with a low cost entry point into the digital world of the Laundry industry, without the cost or complexity of enterprise level ERP alternatives
- The mobile first responsive design ensures accessibility across all devices, in any location which directly addresses the 96% of users who rely on their smartphones
- The automated workflow engine reduces the need for manual human interventions, which addresses the 45% of users who often experience delays in their laundry orders

5.2.3 Theoretical Contributions

The TAM survey results validate Davis's (1989) model in this specific SME context. The strong Perceived Ease of Use score (4.90) traces directly back to the mobile-first design decision, which itself traces back to the 96% smartphone-usage finding from Phase 1 which is an end-to-end, evidence-linked chain from user research through design to measured acceptance.

5.3 Limitations

As much as the Laundrify system addresses most issues addressed in Capstone 1, it still bears its own limitations as stated below:

- Small and single demographic sample, which consists of 20 university hostel residents for both the Phase 1 requirements survey and the Phase 3 TAM evaluation. The findings may not generalise to non-student laundry customers or to laundry shop business owners or the staff themselves
- Due to Render's unreliable environment variable injection, hardcoded credentials were integrated which forced API keys and the database connection string to be embedded

directly in the backend source code as a temporary solution. This is a genuine, currently unresolved security risk

- Brevo's free tier of 300 emails per day and Render's free-tier resource limits are adequate for demonstration purposes but would need proper upgrading in time for a real SME-scale usage
- The TAM evaluation measured first impressions acceptance after a single interaction with Laundrify, not sustained usage over time and evaluated the system primarily from the customers side of view. The admin and staff experience was not studied, and was verified functionally but not evaluated for perceived usability by any actual shop operators
- Testing was conducted by a single tester, which is the individual developer of Laundrify. This may introduce tester bias. Multiple independent testers would be a huge helping hand in strengthening the reliability of the findings

5.4 Future Works

The future of Laundrify highly depends on its future works and upscaling works which are stated below:

- Highest priority is to resolve the credential handling limitation by either fixing Render's environment variable injection issue or to adopt a dedicated secrets manager. Not only that, the backend system of Laundrify could be totally migrated to a different platform with reliable environment variable support, which may cost more.
- Expand the evaluation to a larger and more demographically diverse range of participant pool which includes both consumers, which are the hostel residents and laundry shop owners and staff
- Integrate an actual payment gateway with either Touch n Go or generic FPX Express Pay (Maybank, Public Bank etc etc) to support online pre-payment which reduces the reliance on manual cash reconciliation
- Monitor Brevo usage against the free tier 300 emails per day plan, and come up with a different strategy by upgrading path or fallback provider as usage grows. If the Render's issue is fixed or backend is migrated to a different platform, Gmail SMTP may be supported which is more reliable without any limits.
- Expand testing to include multiple independent testers to reduce the potential tester bias

5.5 Lessons Learned

Throughout the development of Laundrify, there were many lessons that came to me in which I will implement in my future projects. Some of the lessons learned are as stated below:

Email Provider Selection

Gmail SMTP works locally but may be blocked by generic free tier hosting platforms. API-based providers such as Brevo are more reliable for production deployments. The lesson

learned is to test email functionality in production environment early and not just locally to adjust and adapt to environmental issues

Environment Variable Management

Render's unreliable environment variable injection issues highlighted the importance of proper configuration management in a large-scale app development such as Laundrify. A secrets management solution that works regardless of the hosting platform should be implemented and hardcoding credentials should not be the first choice due to potential security risks.

Testing Against Running Systems

Several defects were only discovered once the application was deployed and viewed as an end user would use it. It would be recommended to always test against the running application and not just the source code alone

Free Tier Limitations

Since free tier platforms often have hidden limitations which is only discovered during deployment, it would be recommended to read the fine print and do proper research on these free tier limitations thoroughly before selecting the services.

5.6 Threats to Validity

The study's validity and reliability present several critical methodological limitations that must be addressed in future research. Regarding internal validity, the TAM survey was administered to the same participants who completed the Phase 1 survey, creating a potential response bias that could be mitigated by utilizing independent participant samples. External validity is constrained by the small sample size (N=20) comprised exclusively of university hostel residents, which limits the generalizability of the findings to a broader, more diverse laundry service user base. While the construct validity is supported by the well-validated nature of the TAM instrument, the reliance on self-administered and self-reported surveys introduces inherent subjectivity. Finally, the study's reliability is restricted because the Black Box Testing was conducted by a single tester who was also the developer, a factor that may introduce tester bias into the evaluation process.

5.7 Final Conclusion

This web app, Laundrify demonstrates that the operational transparency gap which was identified in Chapter 1, where 70% of the surveyed users who must physically check on their order can be overcome with a low-cost, free tier deployable web application which is built on evidence gathered directly from the target users.

The project's contributions are broken down into three segments

1. Technical: A timer based automated workflow engine that removes the need for staff to perform manual status tracking which directly addresses the 92% demand for an automatic time estimation system
2. Practical: A documented, evidence-based resolution to a common free-tier hosting issue, SMTP blocking, evaluating different providers before settling on Brevo API. This is a reusable lesson that other students or SME developers can depend on
3. Theoretical: The validation of the Technology Acceptance Model (TAM) in an SME context, with a direct evidence chain from Phase 1 findings, through Phase 2 features to Phase 3 outcomes.

The Laundrify project also showcases the importance of testing against a running system rather than just trusting and assuming based on static code review alone. Several errors were only discovered once the application was exercised as how an end-user would view it.

All 11 test cases reached a passed verification, which demonstrates the system's functional reliability across every tested module, including payment processing under admin override, customer self-payment, and cross-customer authorisation scenarios. This, combined with the strong TAM results and a fully deployed and accessible system, makes Laundrify a credible, evidence based contribution to the digital transformation of micro SMEs in the laundry industry.

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Appendix

Appendix A: Phase 1 Survey Questionnaire and Results

Survey Questionnaire

Section A: Participant Background

Table A.1: Phase 1 Survey Questionnaire - Section A (Participant Background)

Question	Justification
What is your current status? (student / staff / hostel resident)	Categorises the user base and checks whether usage patterns differ by residency status
How often do you utilize laundry services?	Establishes the volume of traffic the system must handle
What type of device do you normally use?	Justifies the need for responsive web design

Section B: Current Experience with Laundry Services

Table A.2: Phase 1 Survey Questionnaire - Section B (Current Experience)

Question	Justification
How do you currently check if your laundry is ready for pickup?	Evaluates the "Transparency Gap"; high physical-checking rates justify a remote status dashboard
Have you ever experienced lost items during laundry services?	Assesses reliability of current tracking; justifies digital order tracking
On a scale of 1-5, how difficult is it to know the exact status of your laundry?	Provides a baseline dissatisfaction score
How often do you find that your laundry is not ready at the promised time?	Validates the need for automated timing

Section C: Feature Preferences and Expectations

Table A.3: Phase 1 Survey Questionnaire - Section C (Feature Preferences)

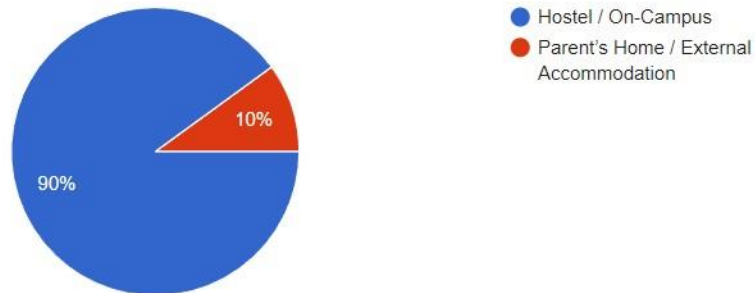
Question	Justification
Which feature would you value the most in a laundry application?	Identifies and prioritises key development requirements
How useful would you find a feature that automatically estimates your pickup time?	Validates demand for the automated timer feature
How important is it for you to rate the quality of the laundry service after completion?	Justifies the customer feedback module
What is your preferred notification method?	Identifies communication preference (60% email)

Phase 1 Survey Results Charts

What is your current residential status?

20 responses

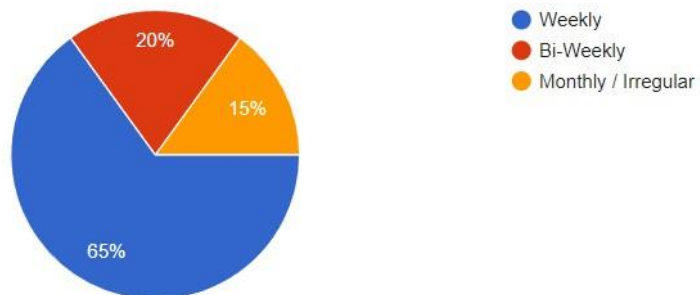
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How frequently do you utilize laundry services?

20 responses

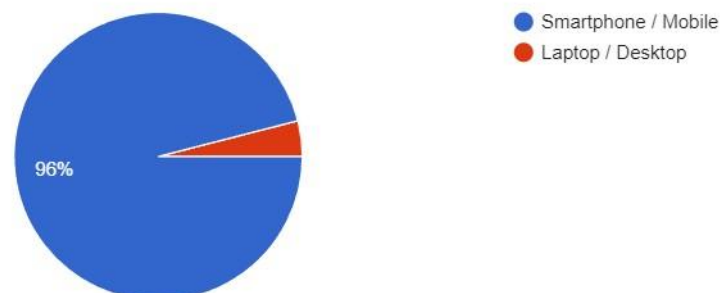
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What type of device do you normally use?

20 responses

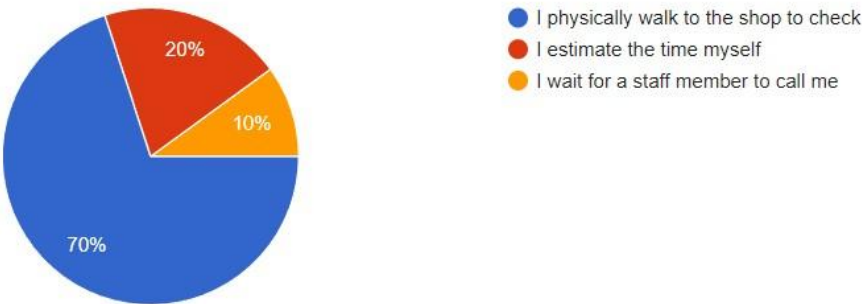
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How do you currently check if your laundry is ready for pickup?

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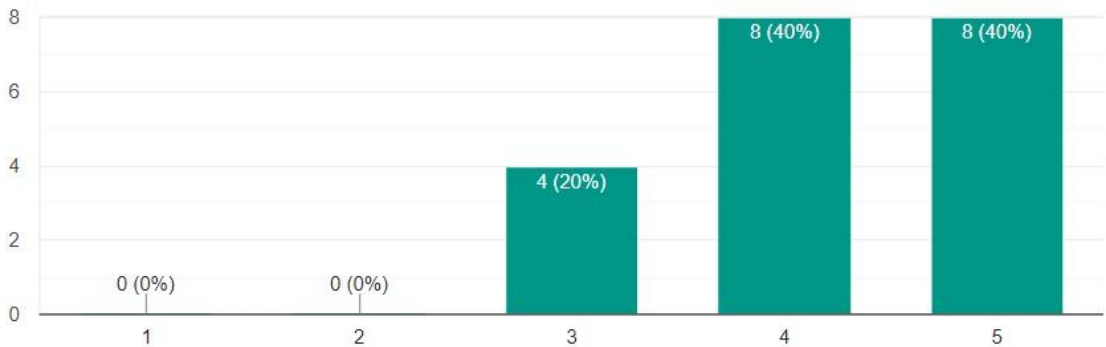
20 responses



On a scale of 1-5, how difficult is it to know the exact status of your laundry?

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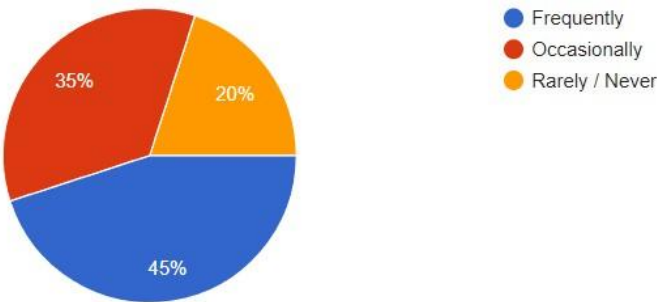
20 responses



How often do you find that your laundry is not ready at the promised time?

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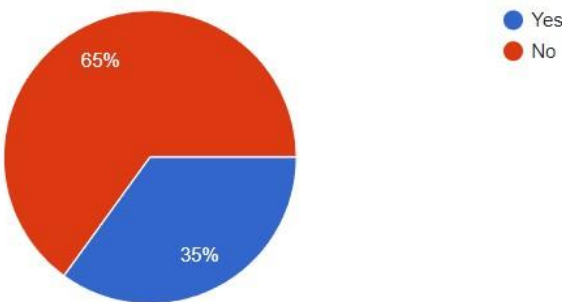
20 responses



Have you ever experienced item loss or mix-ups with your laundry

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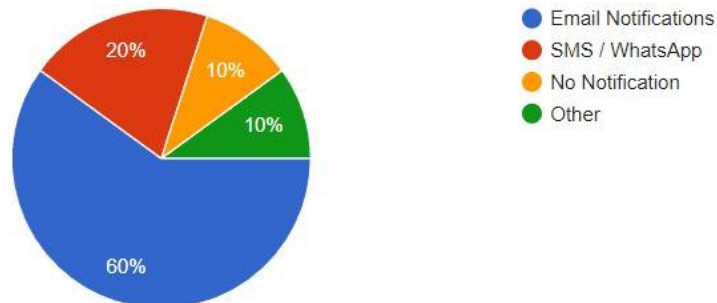
20 responses



What is your preferred method for receiving "Ready for Pickup" alerts

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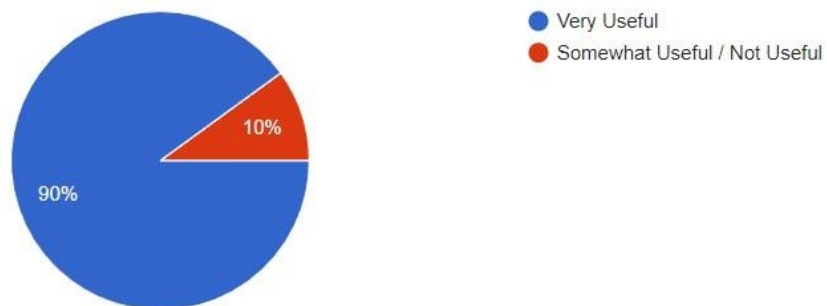
20 responses



How useful would a feature that automatically estimates your completion time be?

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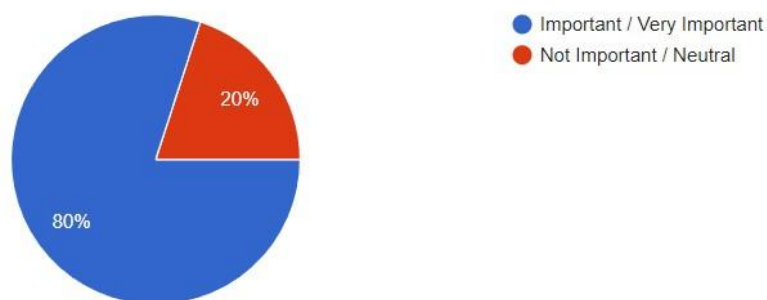
20 responses



How important is it for you to be able to rate the quality of the laundry service?

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20 responses



Appendix B: TAM Survey Instrument

This section contains the full 18-item Technology Acceptance Model survey used in Phase 3 evaluation.

Perceived Usefulness (PU)

Table B.2: Perceived Usefulness (PU) Items

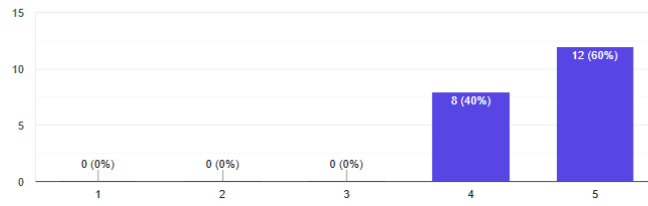
Number	Question
1	The system helps me track my laundry orders easily.
2	The system reduces the time spent on laundry management.
3	The system improves operational efficiency.
4	The system makes it easier to manage payments.
5	The system provides useful analytics and reports.
6	Overall, the system is useful for laundry management.



The system improves operational efficiency

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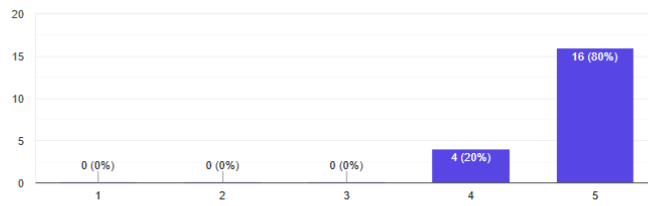
20 responses



The system makes it easier to manage payments

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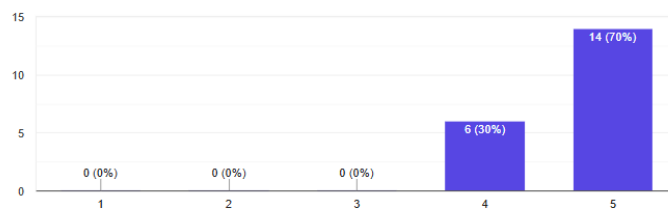
20 responses



The system provides useful analytics and reports

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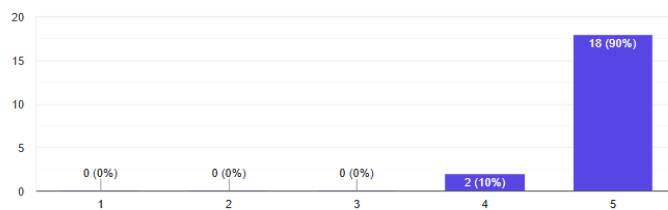
20 responses



Overall, the system is useful for laundry management

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20 responses



Perceived Ease of Use (PEOU)

Table B.2: Perceived Ease of Use (PEOU) Items

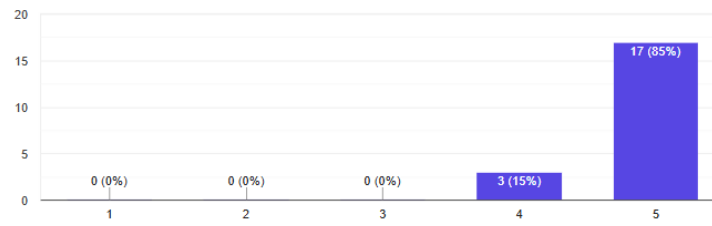
Number	Question
7	The system interface is easy to understand.
8	The system is easy to navigate.
9	Learning to use the system is easy.
10	The system is responsive and fast.
11	The system is well-organized.
12	Overall, the system is easy to use.



Learning to use the system is easy

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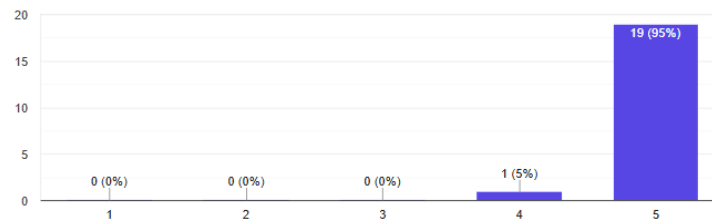
20 responses



The system is responsive and fast

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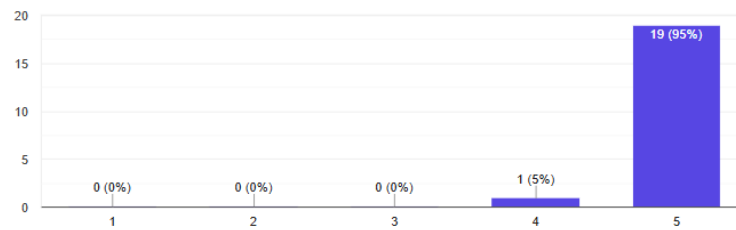
20 responses



The system is well-organized

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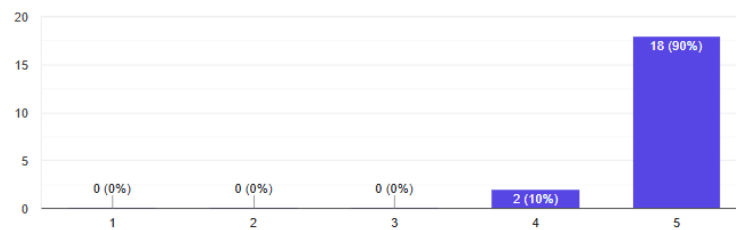
20 responses



Overall, the system is easy to use

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20 responses

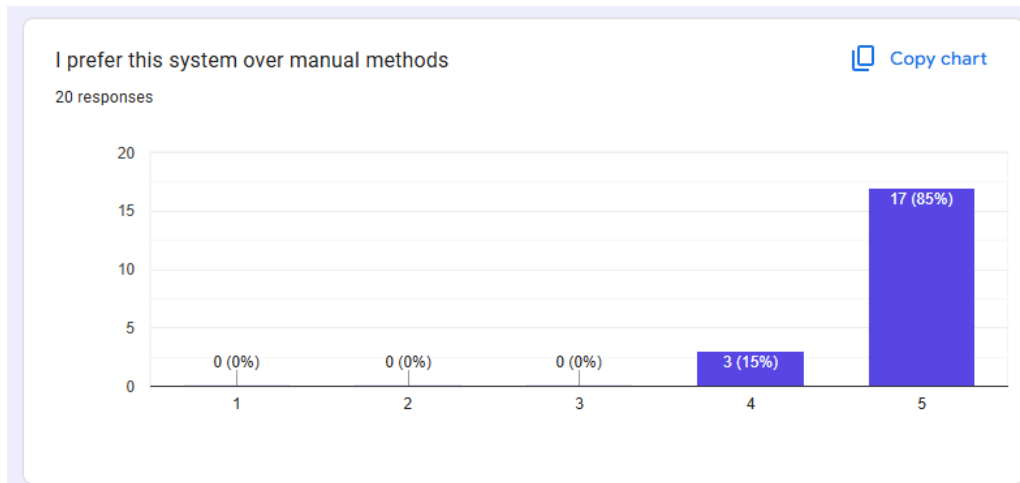


Attitude Toward Using (ATU)

Table B.3: Attitude Toward Using (ATU) Items

Number	Question
13	I find the system enjoyable to use.
14	I would recommend this system to others.
15	I prefer this system over manual methods.



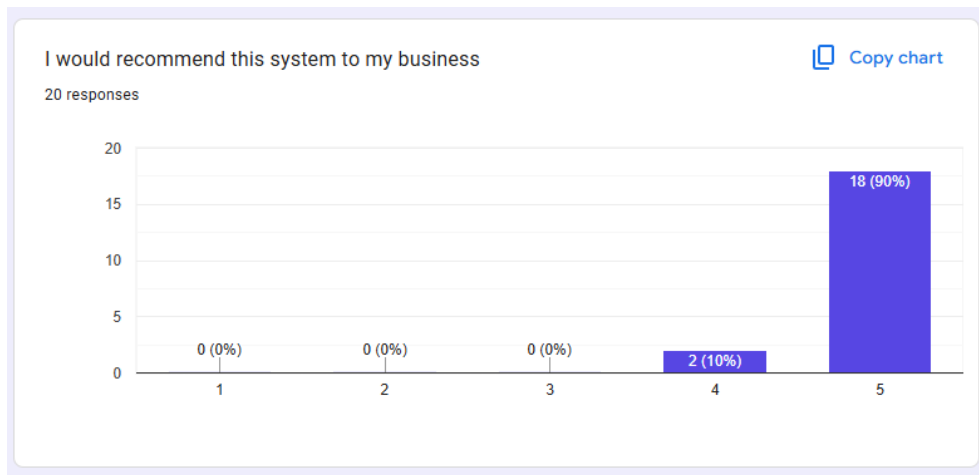


Behavioural Intention (BI)

Table B.4: Behavioural Intention (BI) Items

Number	Question
16	I intend to continue using this system.
17	I would use this system regularly.
18	I would recommend this system to my business.





Disclaimer: All items were rated on a 5-point Likert scale (1 = strongly disagree and 5 = strongly agree)

Appendix C: TAM Survey Raw Response Data (N=20)

This section contains the complete raw response distributions for all 18 TAM items.

Table C.1: TAM Survey Raw Response Data (N=20)

Construct	Item	1	2	3	4	5	Mean
PU	Helps track orders easily	0	0	0	6	14	4.70
PU	Reduces time spent on laundry management	0	0	0	3	17	4.85
PU	Improves operational efficiency	0	0	0	8	12	4.60
PU	Makes it easier to manage payments	0	0	0	4	16	4.80
PU	Provides useful analytics and reports	0	0	0	6	14	4.70
PU	Overall, the system is useful	0	0	0	2	18	4.90
PEOU	Interface is easy to understand	0	0	0	3	17	4.85
PEOU	Easy to navigate	0	0	0	2	18	4.90
PEOU	Learning to use is easy	0	0	0	3	17	4.85

PEOU	Responsive and fast	0	0	0	1	19	4.95
PEOU	Well-organized	0	0	0	1	19	4.95
PEOU	Overall, easy to use	0	0	0	2	18	4.90
ATU	Enjoyable to use	0	0	0	4	16	4.80
ATU	Would recommend to others	0	0	0	3	17	4.85
ATU	Prefer over manual methods	0	0	0	3	17	4.85
BI	Intend to continue using	0	0	0	2	18	4.90
BI	Would use regularly	0	0	0	5	15	4.75
BI	Would recommend to my business	0	0	0	2	18	4.90

Appendix D: Black Box Test Log

This section contains the complete Black Box Testing log from Sprint 3 development.

Table D.1: Black Box Test Log

Module	Test Case	Expected Result	Actual Result	Status
Authentication	Admin role auto-assignment on registration	Designated admin email receives admin role automatically	Admin role assigned correctly at registration	Pass
Role-Based Access	Customer cannot reach Settings via UI	No Settings link rendered for customer role	Navigation array conditioned on role; entry removed	Pass
Role-Based Access	Customer cannot reach Settings via direct navigation	Redirected to dashboard	Route guard redirects non-admin users	Pass
UI Consistency	Order History icon consistent across pages	Same icon used on all pages	All pages use Calendar icon consistently	Pass
Mobile Responsiveness	Mobile header renders on all pages	Logo + hamburger menu visible on mobile	All pages render mobile header correctly	Pass
Order Management	Order pricing includes all applicable surcharges	Total reflects all charges	All surcharges correctly included	Pass

Payment Processing	Admin can update payment status on any order	Payment status updates regardless of owner	Admin override works correctly	Pass
Payment Processing	Customer can update their own payment status	Customer can pay their own order	userId match allows self-payment	Pass
Payment Processing	Customer cannot update another customer's payment status	Update rejected for non-owned orders	Fix applied and retested — query now scopes to userId: req.user._id, so the update is rejected with a 404 for non-owned orders	Pass
Automation	Order auto-completes after estimated time	Status transitions Processing → Completed	Verified via 10-second interval check	Pass
Notifications	Email sent on auto-completion	Customer receives completion email	Email delivered via Brevo API	Pass

Appendix E: Deployment Information

This section contains the deployment architecture and configuration details.

Table E.1: Deployment Architecture and Configuration Details

Detail	Value
Frontend Platform	Vercel
Frontend URL	https://laundrify-seven.vercel.app
Backend Platform	Render
Database Platform	MongoDB Atlas
Email Provider	Brevo API (formerly Sendinblue)
Email Free Tier	300 emails/day
Framework	React (Frontend) + Node.js/Express (Backend)
Authentication	JWT (JSON Web Tokens)

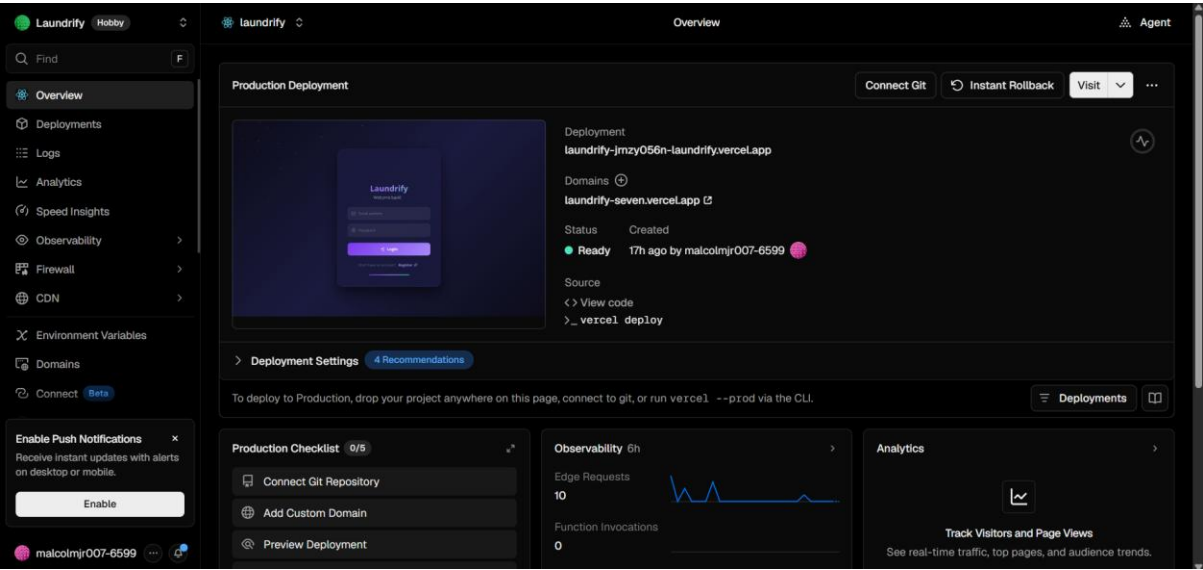


Figure A.1: Vercel Deployment Dashboard

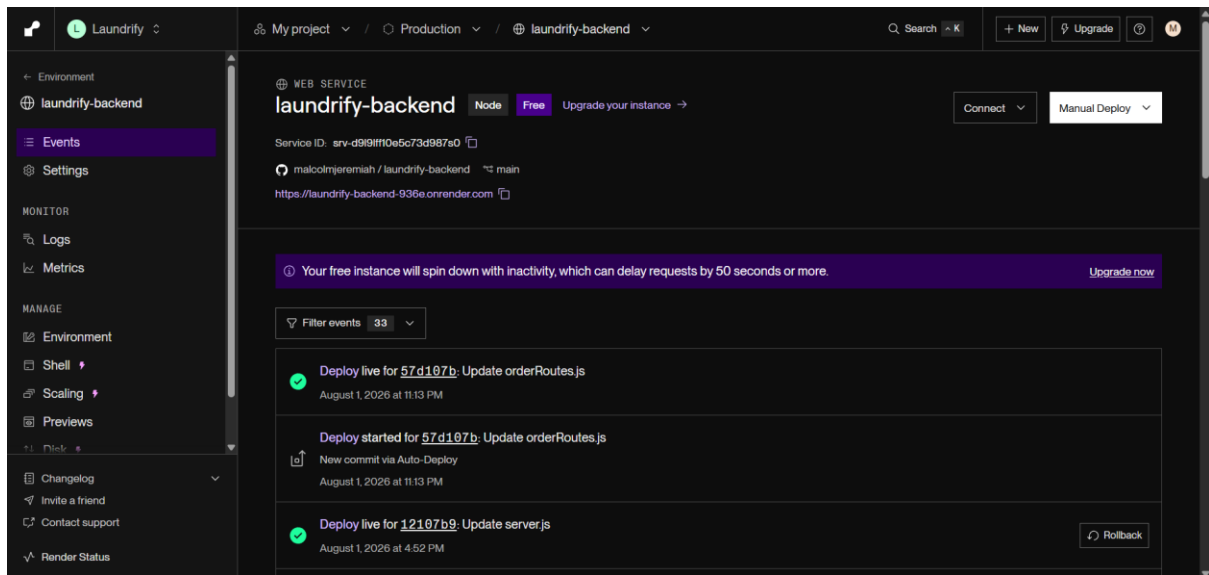


Figure A.2: Render Deployment Dashboard